results/download_era5_data.csv

Extreme Event Analysis Report

1) Descriptive Statistics

Index	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max	skewne...	kurtos...
swh	745128	2.23	1.06	0.41	0.72	0.95	1.47	2.01	2.75	4.28	5.68	11.19	1.33	2.67
mwd	745128	301.28	51.82	0	6	233	289	310	330	350	357	360	-3.50	16.61
pp1d	745128	10.77	2.64	3	5.40	6.40	8.80	10.90	12.60	15	16.60	21.40	0.03	-0.48
wind	745128	7.03	3.27	2	2	2	4.40	6.77	9.37	12.72	14.84	27.12	0.39	-0.49
dwi	745128	205.43	127.39	0	1	5	75	228	331	356	359	360	-0.38	-1.34

2) GEV: swh (all directions)

Shape=0.1189, Loc=7.1752, Scale=1.0197

Return Periods => Return Levels (swh):

T=2: 7.54

T=5: 8.58

T=10: 9.19

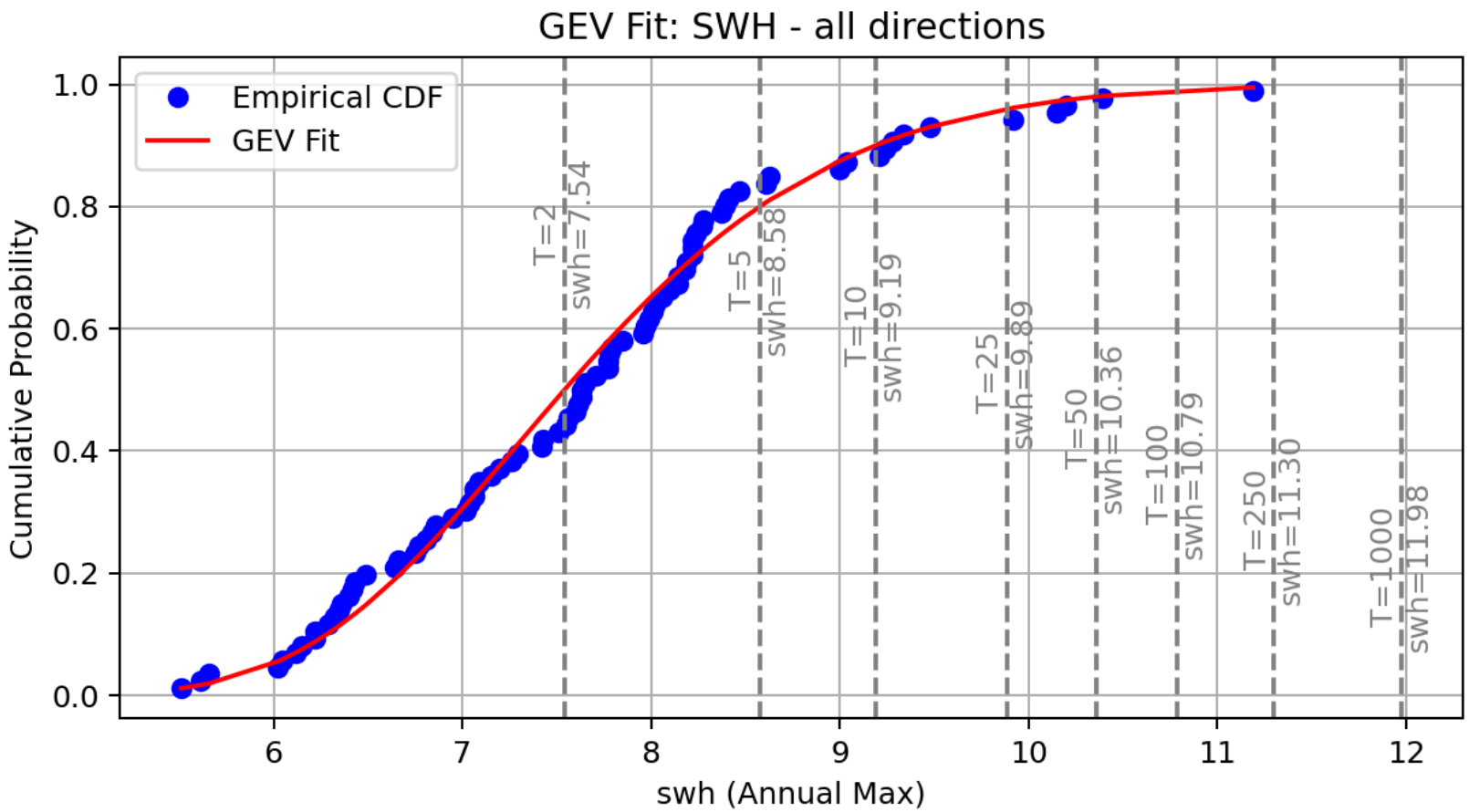
T=25: 9.89

T=50: 10.36

T=100: 10.79

T=250: 11.30

T=1000: 11.98

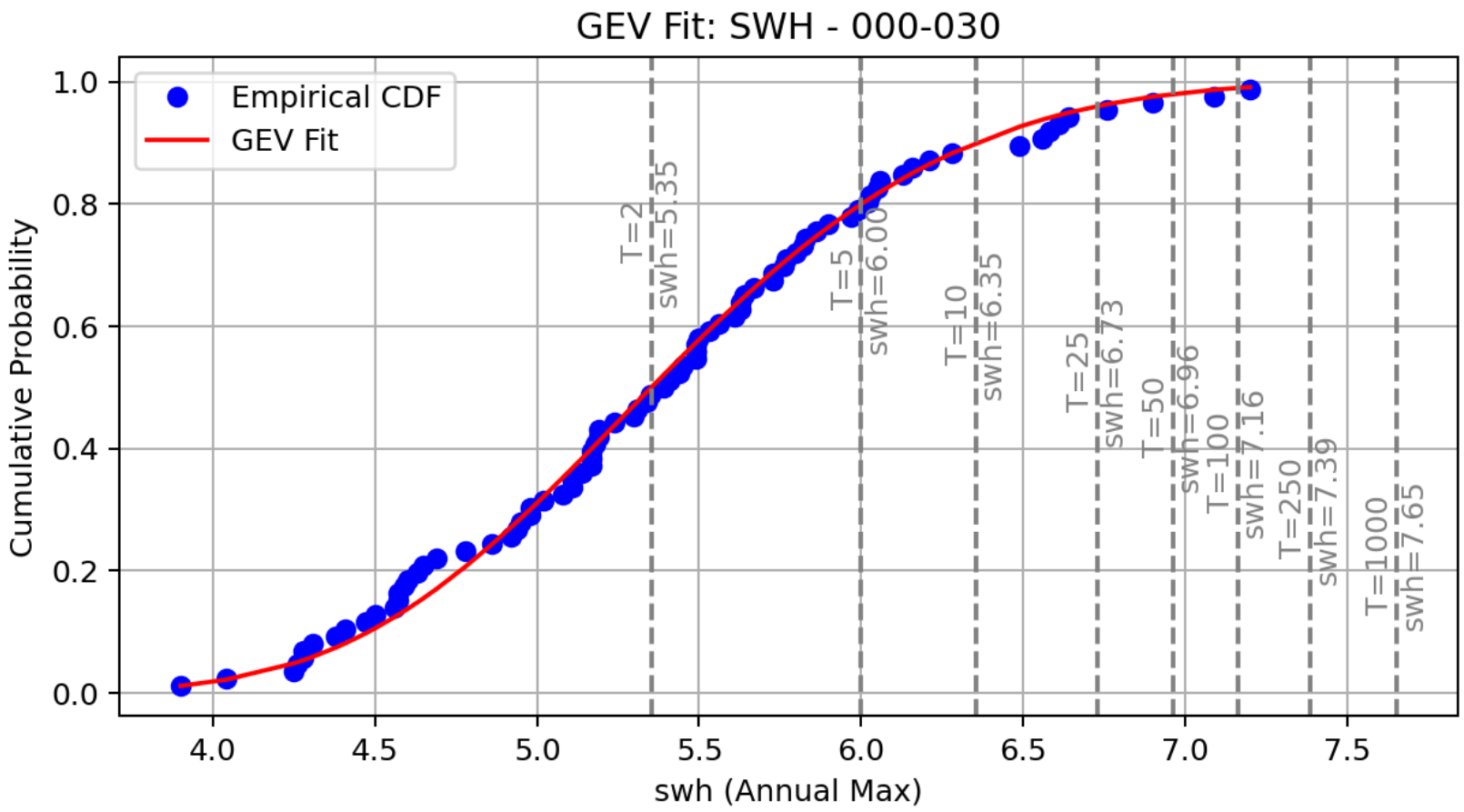

GEV Fit: SWH - all directions
Empirical CDF
GEV Fit
Cumulative Probability
swh (Annual Max)
T=2
swh=7.54
T=5
swh=8.58
T=10
swh=9.19
T=25
swh=9.89
T=50
swh=10.36
T=100
swh=10.79
T=250
swh=11.30
T=1000
swh=11.98

3) GEV: swh by 30° Sectors

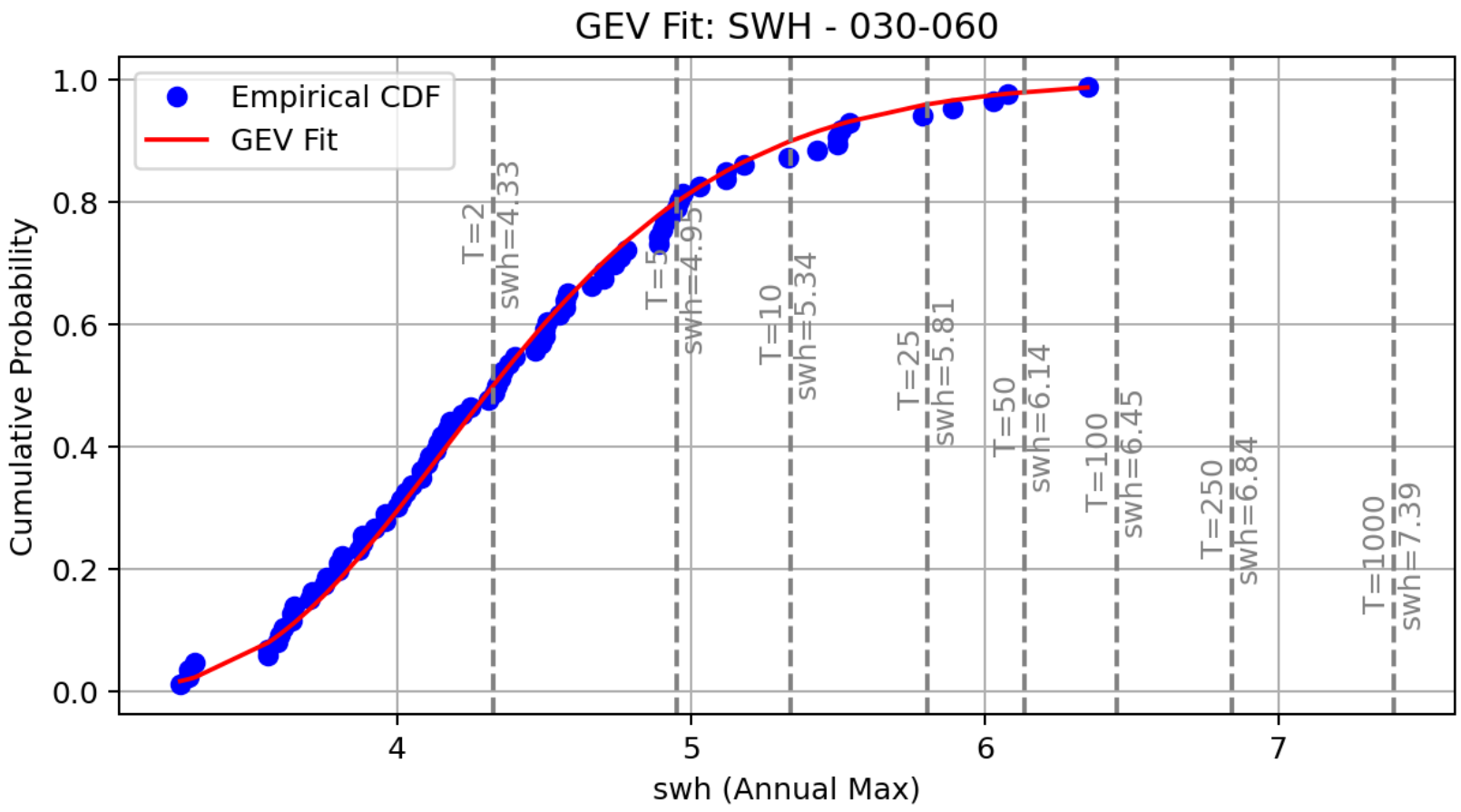
SWH GEV by Sector

Index	shape	loc	scale	RP_2	RP_5	RP_10	RP_25	RP_50	RP_100	RP_250	RP_1000
000-030	0.21	5.11	0.69	5.35	6.00	6.35	6.73	6.96	7.16	7.39	7.65
030-060	0.06	4.12	0.58	4.33	4.95	5.34	5.81	6.14	6.45	6.84	7.39
060-090	0.10	3.76	0.66	3.99	4.67	5.08	5.56	5.89	6.19	6.56	7.06
090-120	0.05	3.77	0.69	4.02	4.76	5.23	5.80	6.21	6.60	7.09	7.79
120-150	0.18	4.21	0.81	4.50	5.27	5.71	6.18	6.47	6.74	7.03	7.40
150-180	0.00	4.92	0.71	5.18	5.98	6.51	7.18	7.67	8.16	8.81	9.78
180-210	-0.02	5.42	0.82	5.72	6.67	7.31	8.14	8.76	9.40	10.25	11.57
210-240	0.03	5.95	0.98	6.30	7.37	8.06	8.91	9.52	10.11	10.87	11.97
240-270	0.12	6.34	1.09	6.73	7.84	8.50	9.24	9.74	10.20	10.74	11.46
270-300	0.18	6.54	1.00	6.89	7.86	8.40	8.98	9.36	9.69	10.07	10.53
300-330	0.17	6.42	1.12	6.82	7.90	8.51	9.19	9.62	10.00	10.44	10.99
330-360	0.08	5.94	0.98	6.29	7.32	7.95	8.69	9.20	9.68	10.27	11.07

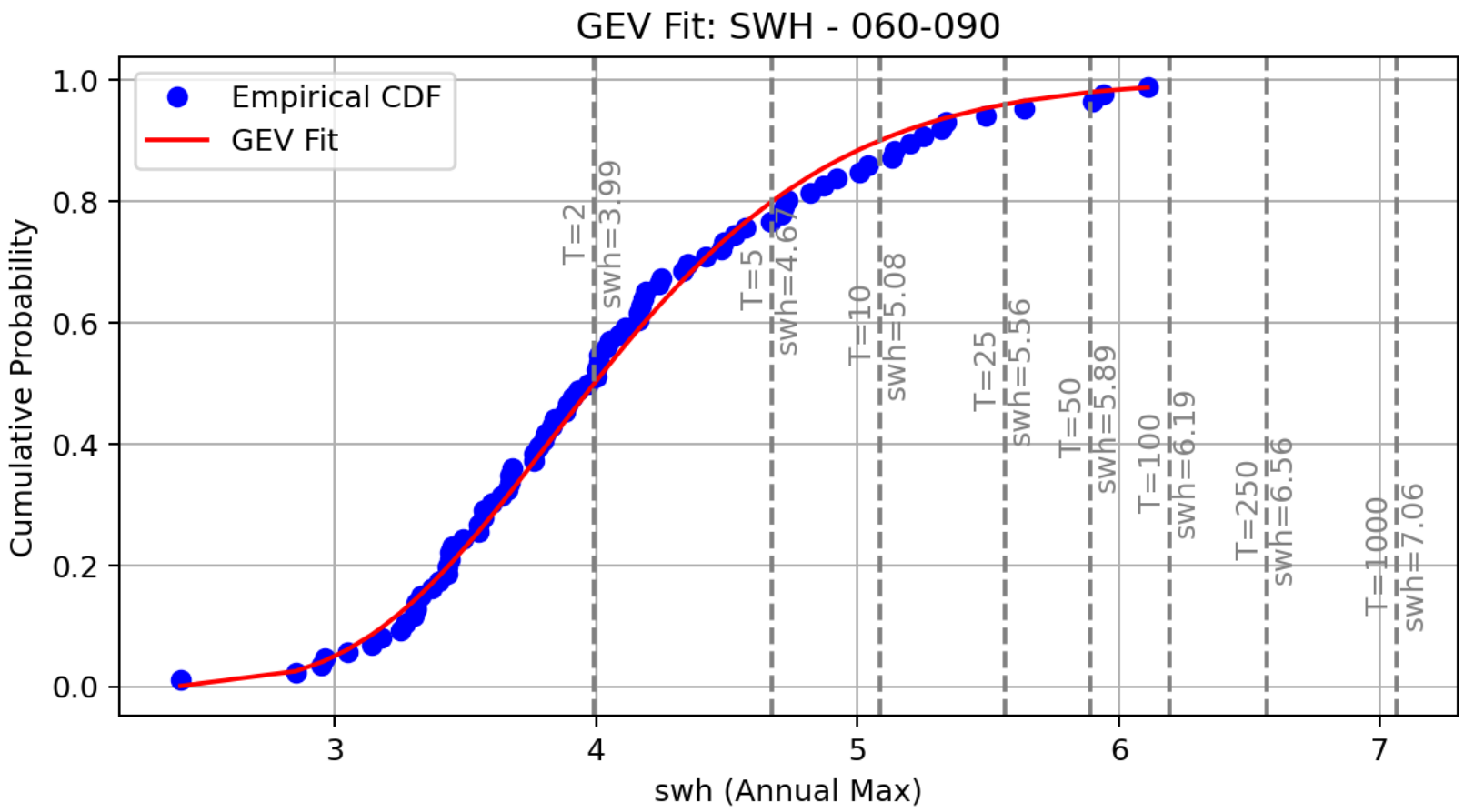
SWH Sector 000-030



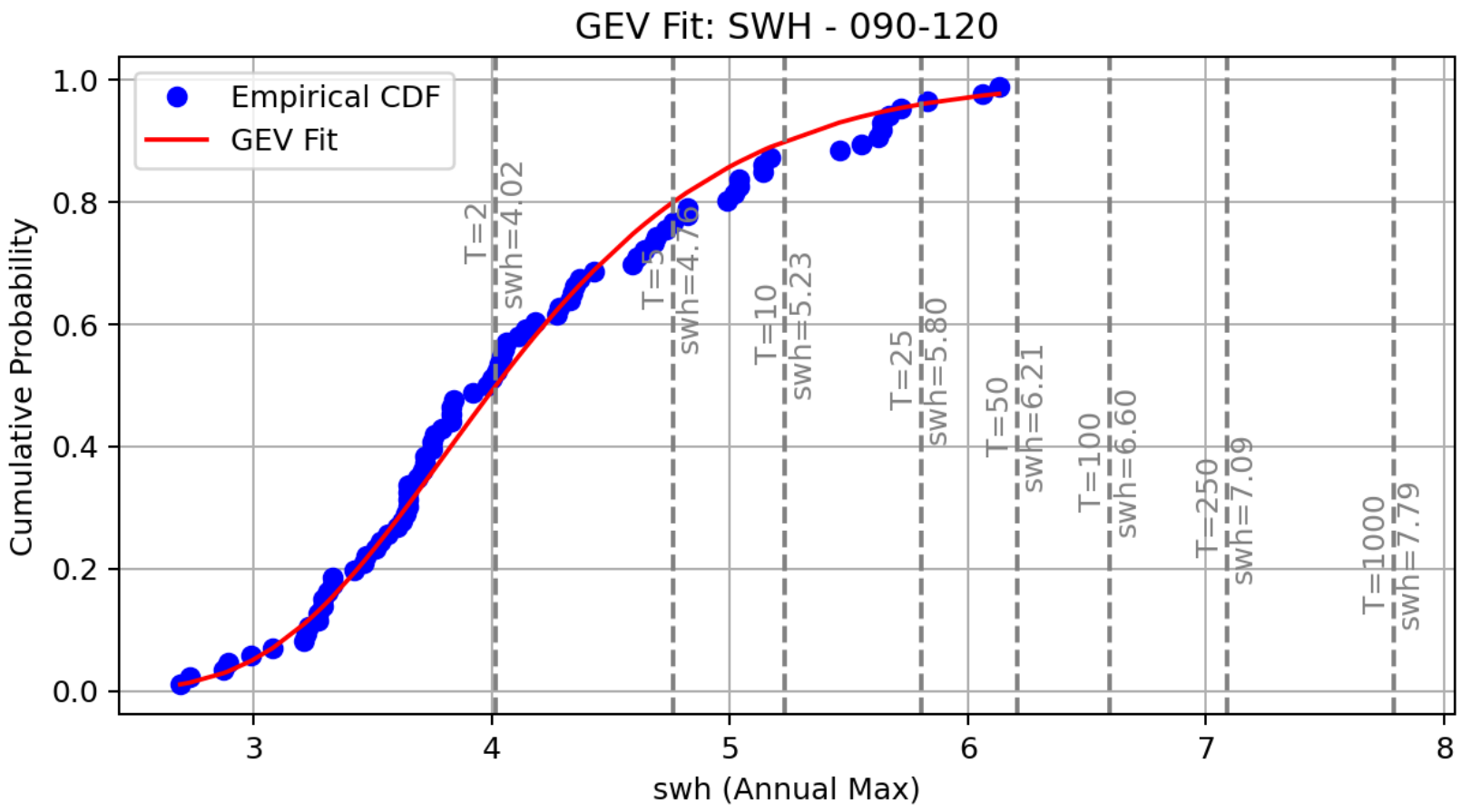
SWH Sector 030-060


GEV Fit: SWH - 030-060
Empirical CDF
GEV Fit
Cumulative Probability
swh (Annual Max)
1.0
0.8
0.6
0.4
0.2
0.0
4
5
6
7
T=2
swh=4.33
T=5
swh=4.95
T=10
swh=5.34
T=25
swh=5.81
T=50
swh=6.14
T=100
swh=6.45
T=250
swh=6.84
T=1000
swh=7.39

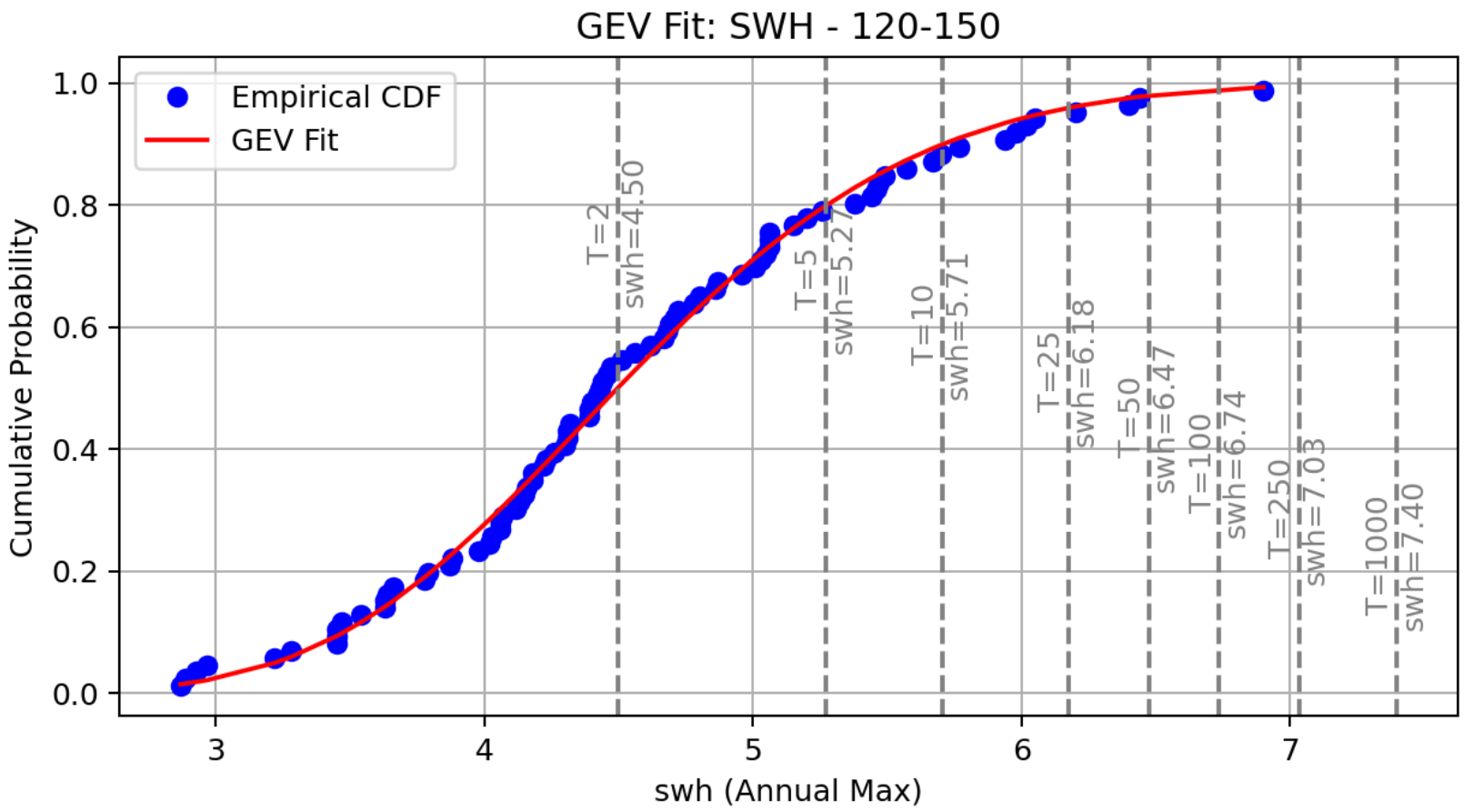
SWH Sector 060-090



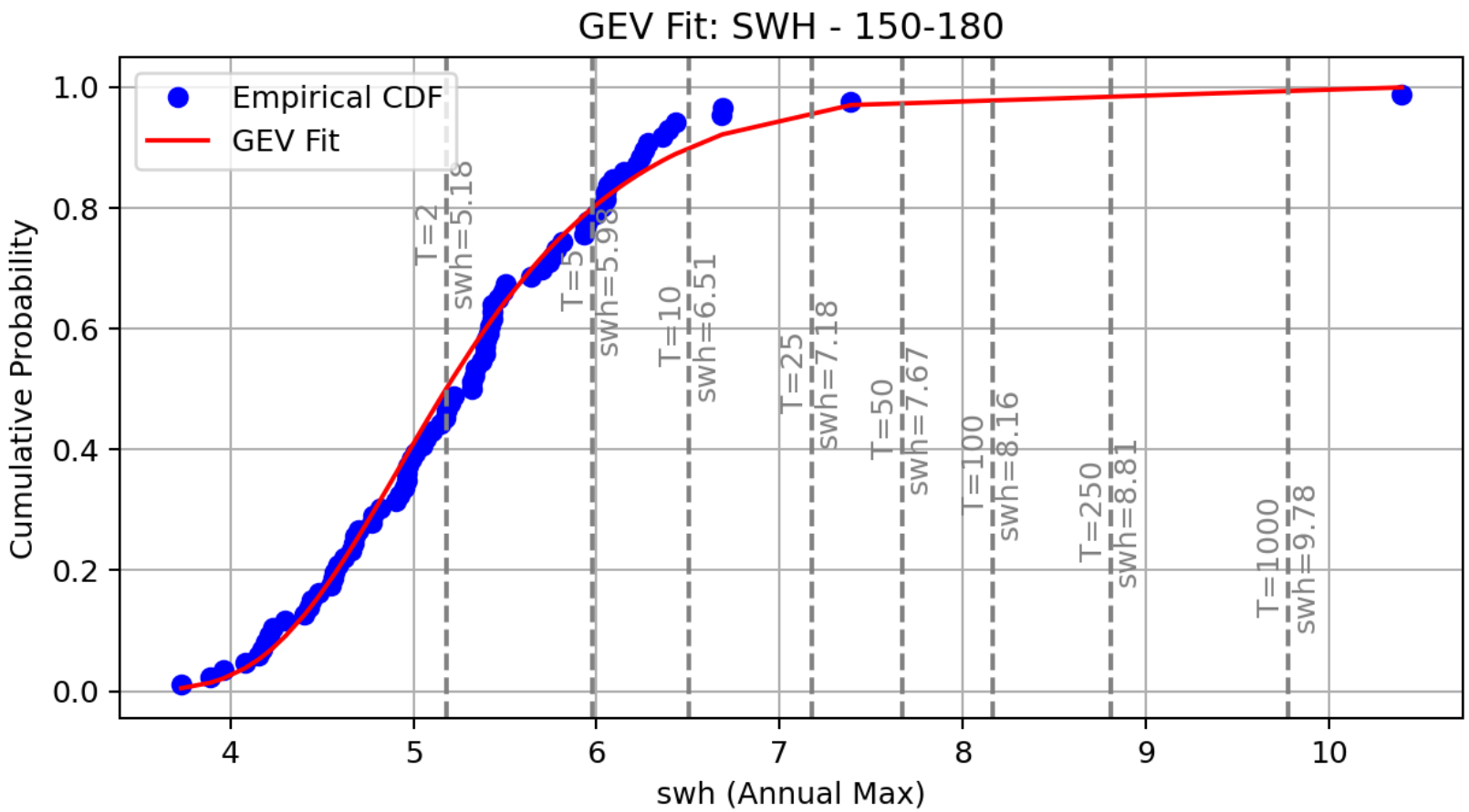
SWH Sector 090-120


GEV Fit: SWH - 090-120
Empirical CDF
GEV Fit
Cumulative Probability
swh (Annual Max)
T=2
swh=4.02
T=5
swh=4.76
T=10
swh=5.23
T=25
swh=5.80
T=50
swh=6.21
T=100
swh=6.60
T=250
swh=7.09
T=1000
swh=7.79

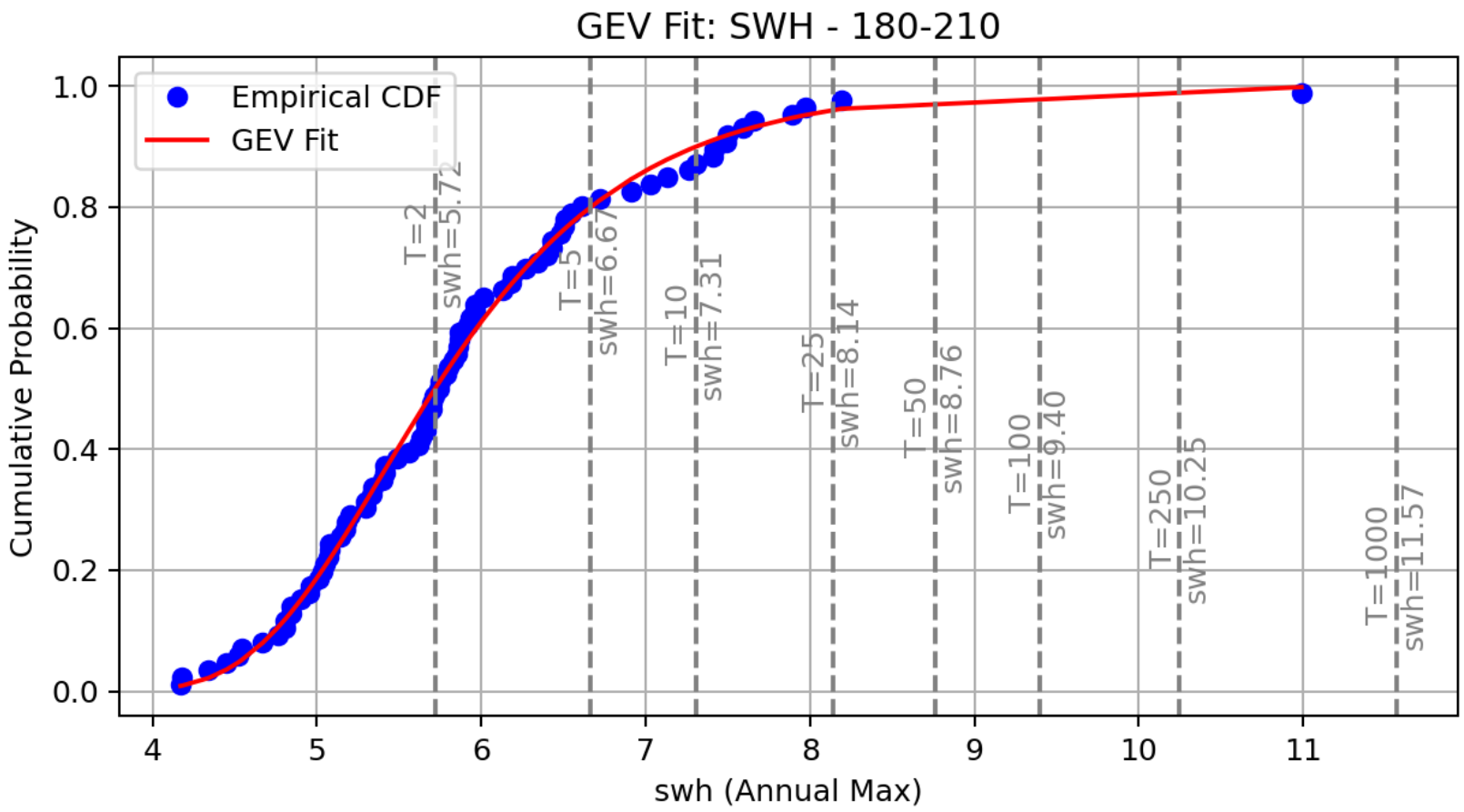
SWH Sector 120-150



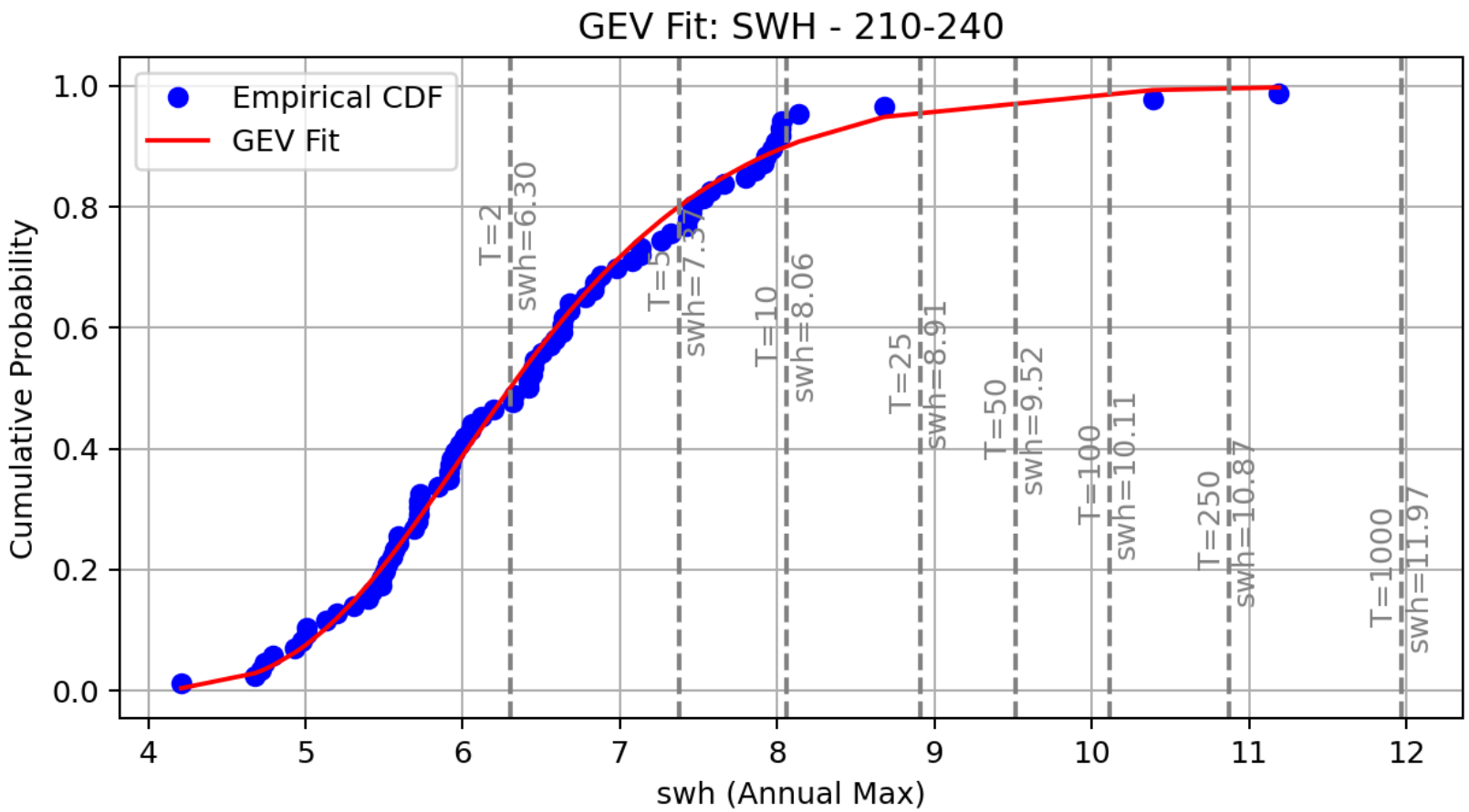
SWH Sector 150-180


GEV Fit: SWH - 150-180
Empirical CDF
GEV Fit
T=2
swh=5.18
T=5
swh=5.98
T=10
swh=6.51
T=25
swh=7.18
T=50
swh=7.67
T=100
swh=8.16
T=250
swh=8.81
T=1000
swh=9.78
Cumulative Probability
swh (Annual Max)
0.0
0.2
0.4
0.6
0.8
1.0
4
5
6
7
8
9
10

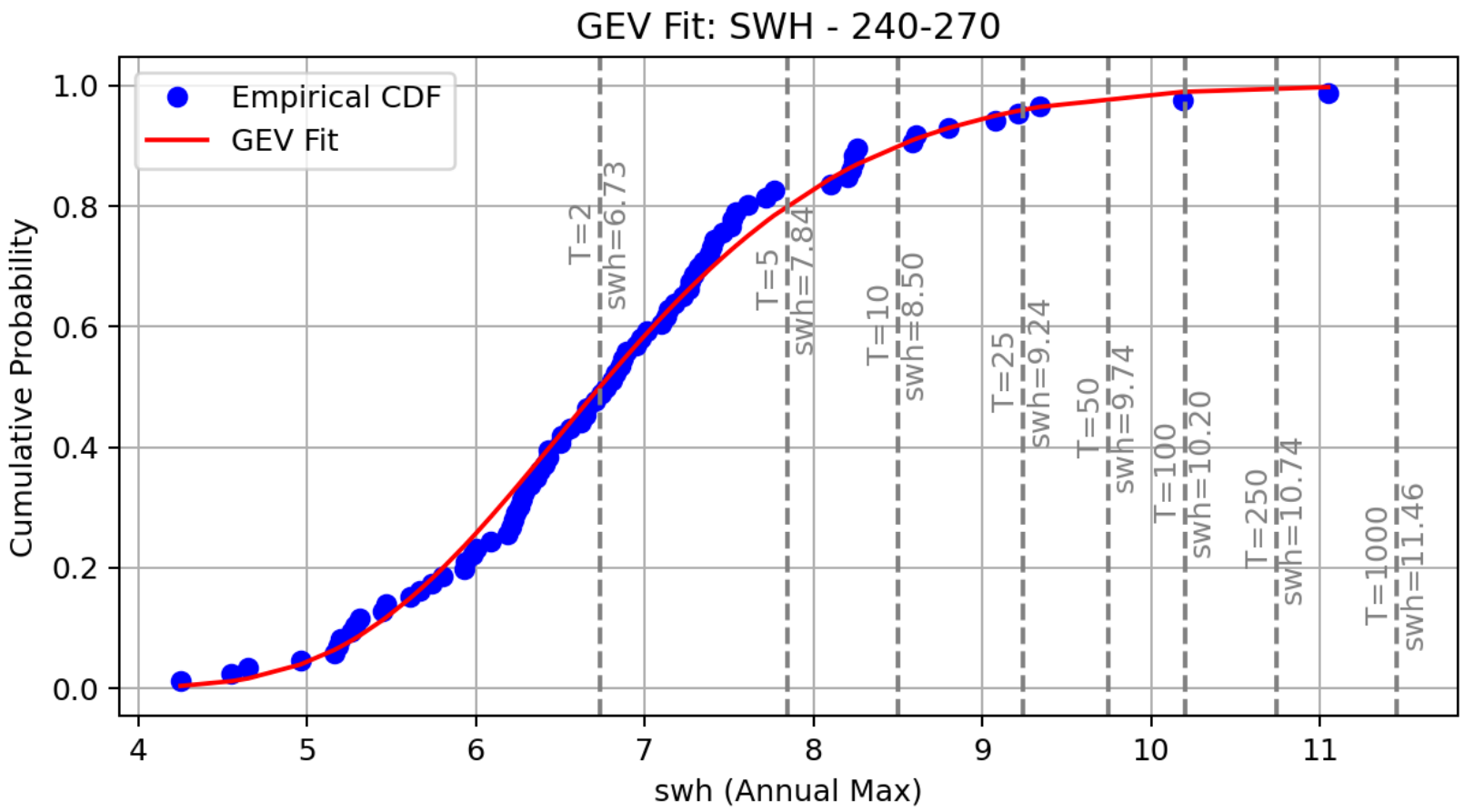
SWH Sector 180-210



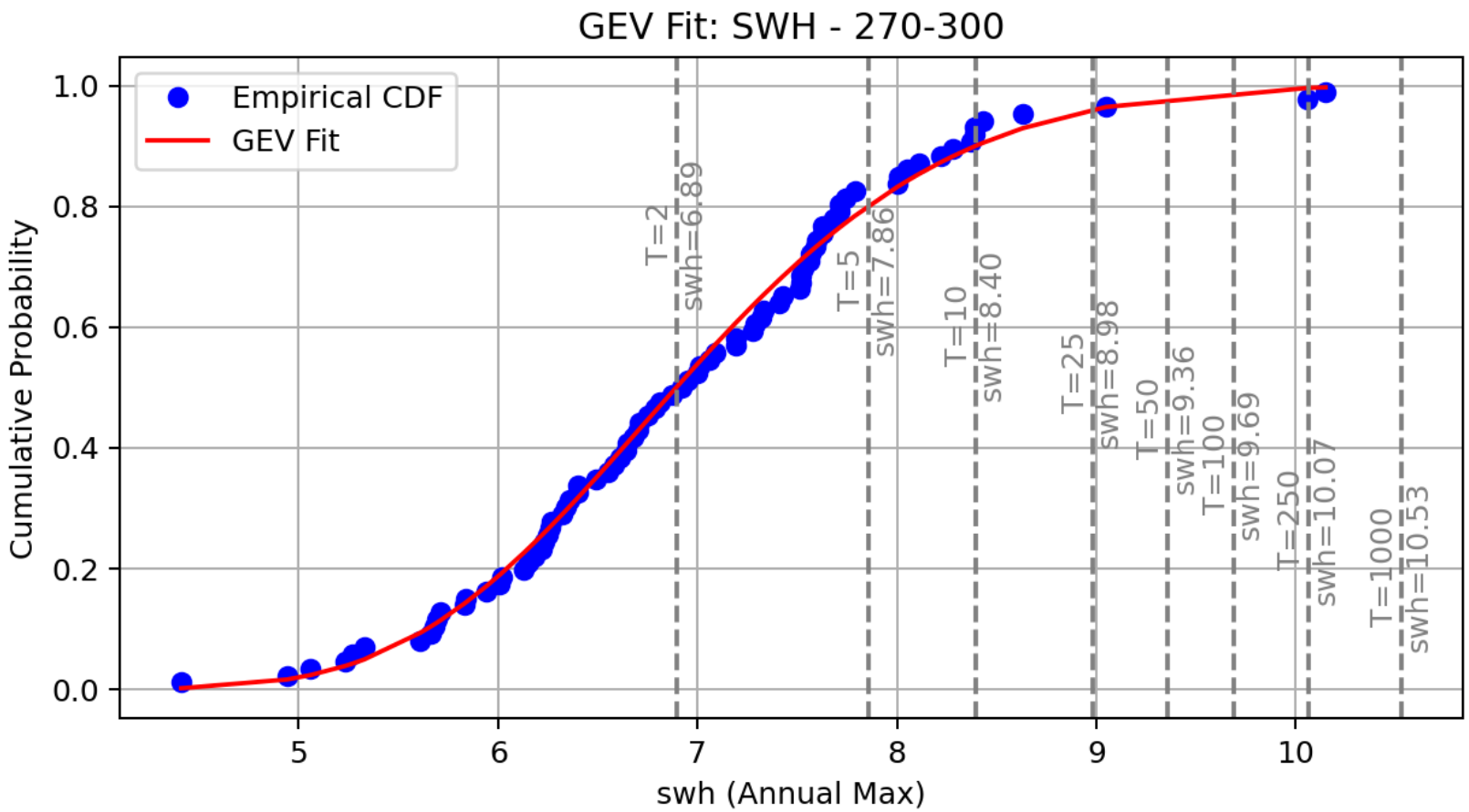
SWH Sector 210-240


GEV Fit: SWH - 210-240
Empirical CDF
GEV Fit
Cumulative Probability
swh (Annual Max)
T=2
swh=6.30
T=5
swh=7.34
T=10
swh=8.06
T=25
swh=8.91
T=50
swh=9.52
T=100
swh=10.11
T=250
swh=10.87
T=1000
swh=11.97
0.0
0.2
0.4
0.6
0.8
1.0
4
5
6
7
8
9
10
11
12

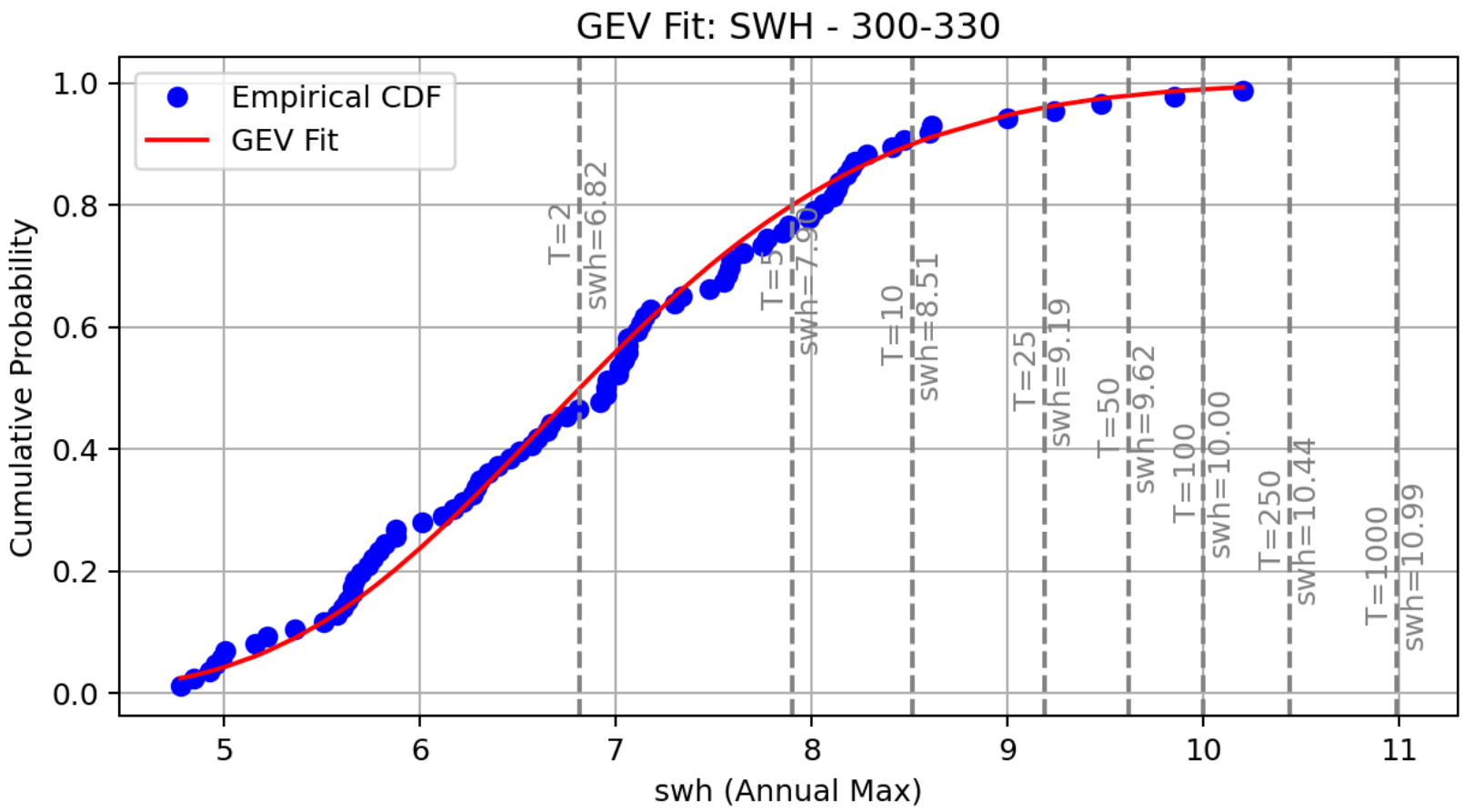
SWH Sector 240-270



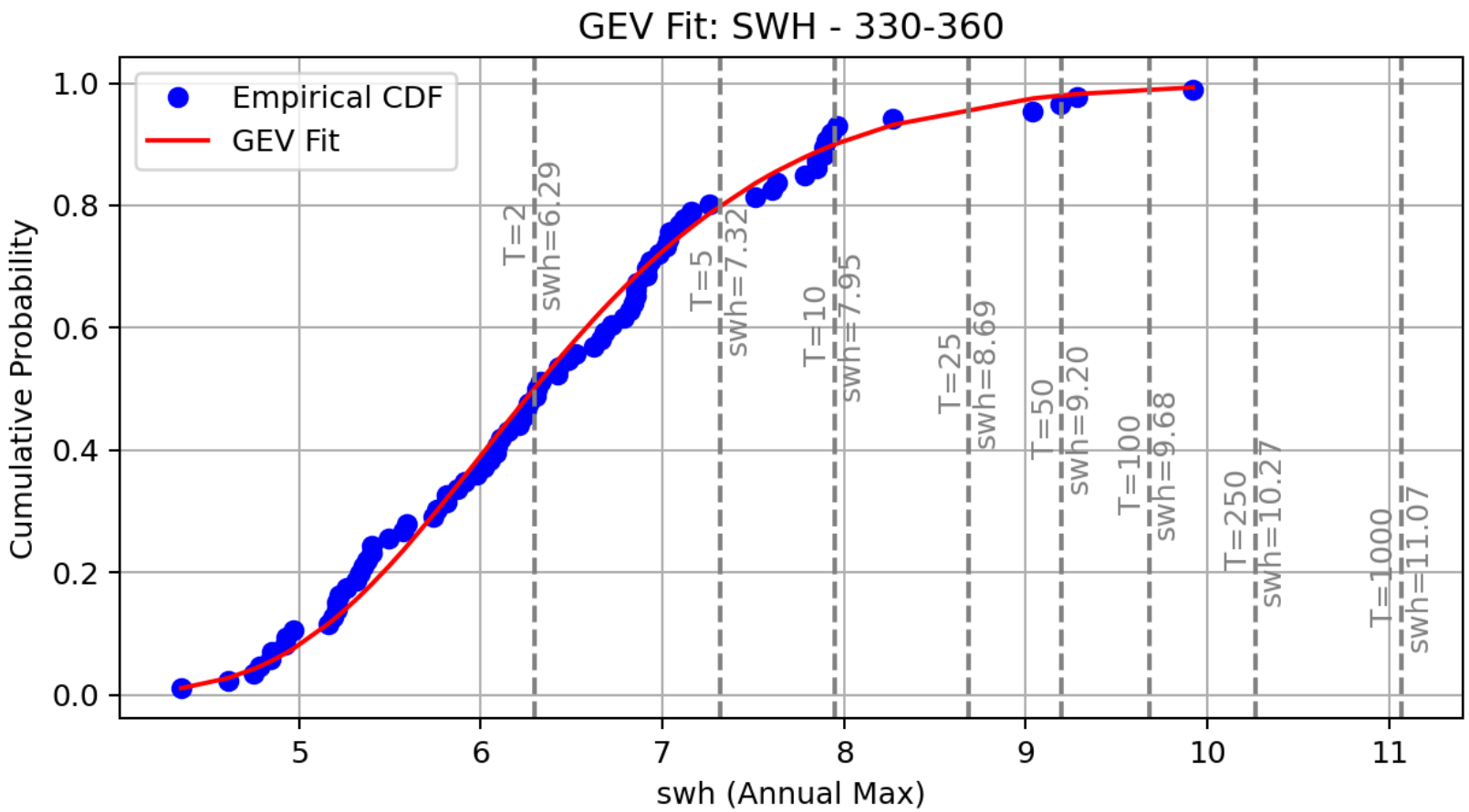
SWH Sector 270-300


GEV Fit: SWH - 270-300
Empirical CDF
GEV Fit
Cumulative Probability
swh (Annual Max)
1.0
0.8
0.6
0.4
0.2
0.0
5
6
7
8
9
10
T=2
swh=6.89
T=5
swh=7.86
T=10
swh=8.40
T=25
swh=8.98
T=50
swh=9.36
T=100
swh=9.69
T=250
swh=10.07
T=1000
swh=10.53

SWH Sector 300-330



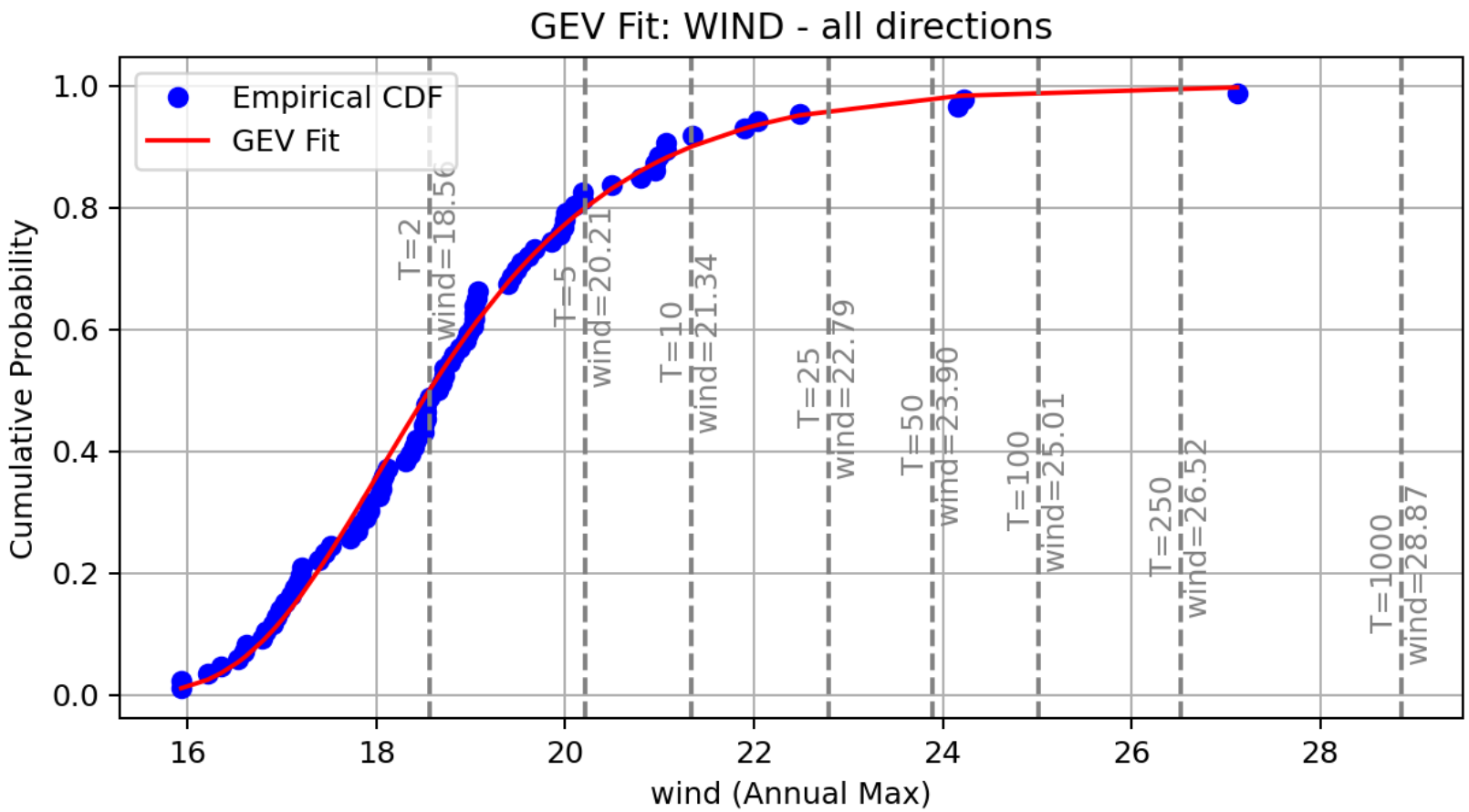
SWH Sector 330-360


GEV Fit: SWH - 330-360
Empirical CDF
GEV Fit
Cumulative Probability
swh (Annual Max)
T=2
swh=6.29
T=5
swh=7.32
T=10
swh=7.95
T=25
swh=8.69
T=50
swh=9.20
T=100
swh=9.68
T=250
swh=10.27
T=1000
swh=11.07
0.0
0.2
0.4
0.6
0.8
1.0
5
6
7
8
9
10
11

4) GEV: wind (all directions)

```
Shape=-0.0281, Loc=18.0381, Scale=1.4203

Return Periods => Return Levels (wind):
  T=2: 18.56
  T=5: 20.21
  T=10: 21.34
  T=25: 22.79
  T=50: 23.90
  T=100: 25.01
  T=250: 26.52
  T=1000: 28.87
```

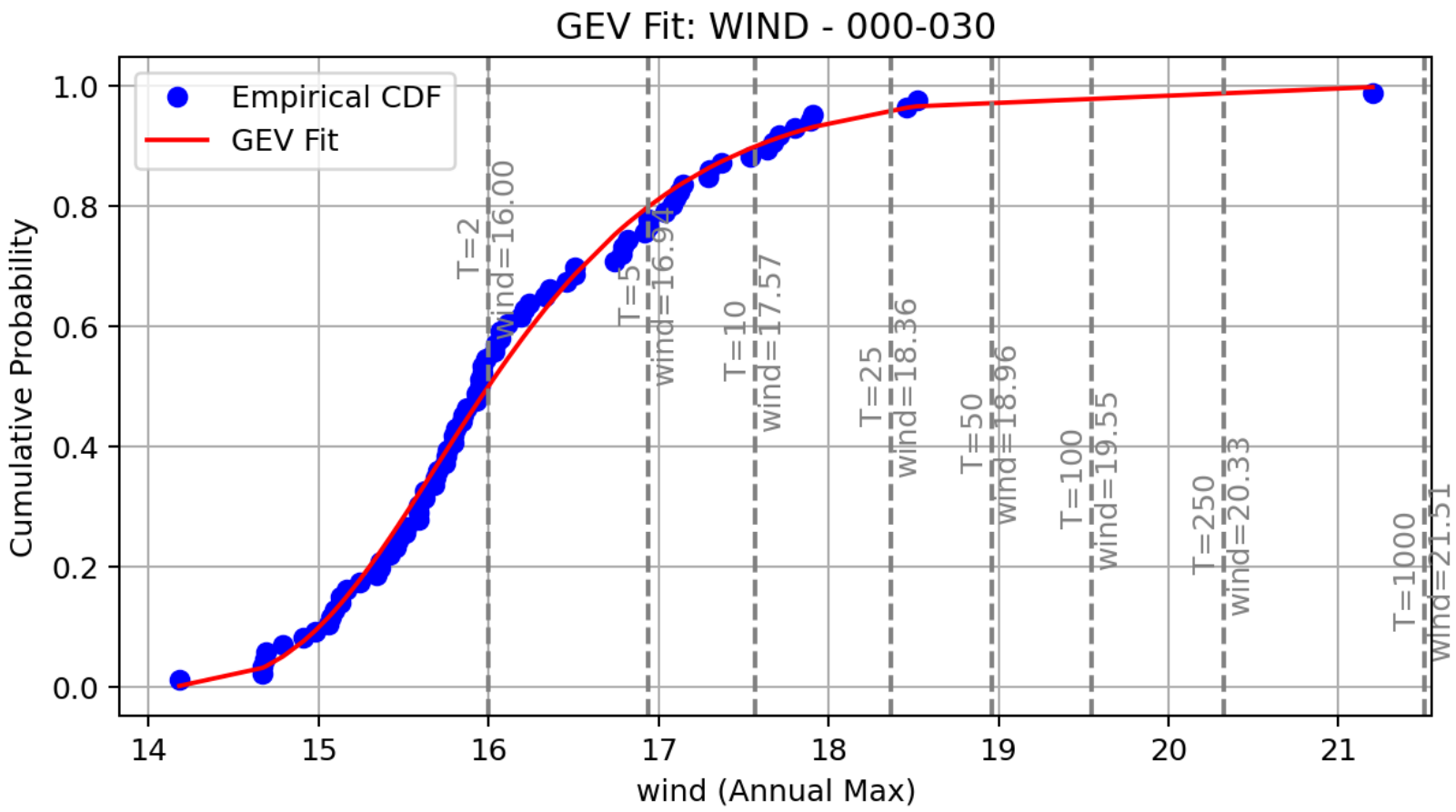

GEV Fit: WIND - all directions
Empirical CDF
GEV Fit
T=2
wind=18.56
T=5
wind=20.21
T=10
wind=21.34
T=25
wind=22.79
T=50
wind=23.90
T=100
wind=25.01
T=250
wind=26.52
T=1000
wind=28.87
Cumulative Probability
wind (Annual Max)
0.0
0.2
0.4
0.6
0.8
1.0
16
18
20
22
24
26
28

5) GEV: wind by 30° Sectors

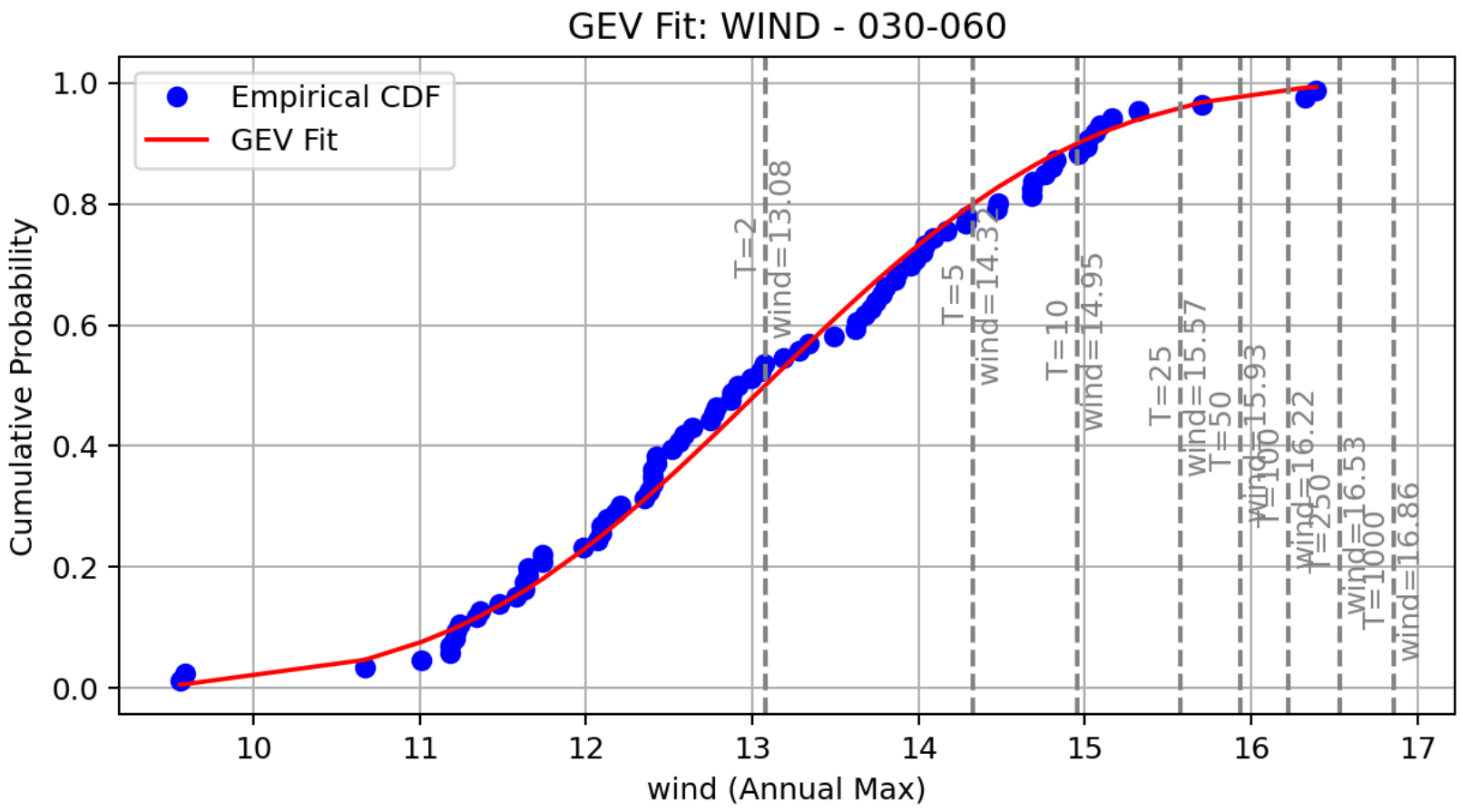
WIND GEV by Sector

Index	shape	loc	scale	RP_2	RP_5	RP_10	RP_25	RP_50	RP_100	RP_250	RP_1000
000-030	-0.00	15.69	0.83	16.00	16.94	17.57	18.36	18.96	19.55	20.33	21.51
030-060	0.29	12.58	1.44	13.08	14.32	14.95	15.57	15.93	16.22	16.53	16.86
060-090	0.22	10.57	1.33	11.03	12.26	12.92	13.62	14.04	14.41	14.81	15.28
090-120	0.07	10.78	1.52	11.33	12.95	13.96	15.16	16.01	16.81	17.80	19.19
120-150	0.21	13.52	2.53	14.41	16.77	18.04	19.39	20.22	20.93	21.72	22.66
150-180	0.05	15.93	1.68	16.54	18.36	19.51	20.90	21.90	22.85	24.06	25.79
180-210	0.08	16.26	1.51	16.80	18.39	19.37	20.52	21.32	22.07	23.00	24.28
210-240	0.06	16.26	1.57	16.83	18.51	19.56	20.81	21.70	22.54	23.59	25.07
240-270	0.17	16.16	1.84	16.81	18.60	19.60	20.70	21.41	22.04	22.75	23.64
270-300	0.13	16.13	1.61	16.71	18.33	19.29	20.36	21.08	21.74	22.51	23.52
300-330	0.09	15.60	1.99	16.32	18.40	19.66	21.14	22.15	23.10	24.26	25.84
330-360	-0.19	15.67	1.13	16.10	17.64	18.85	20.66	22.22	23.99	26.71	31.83

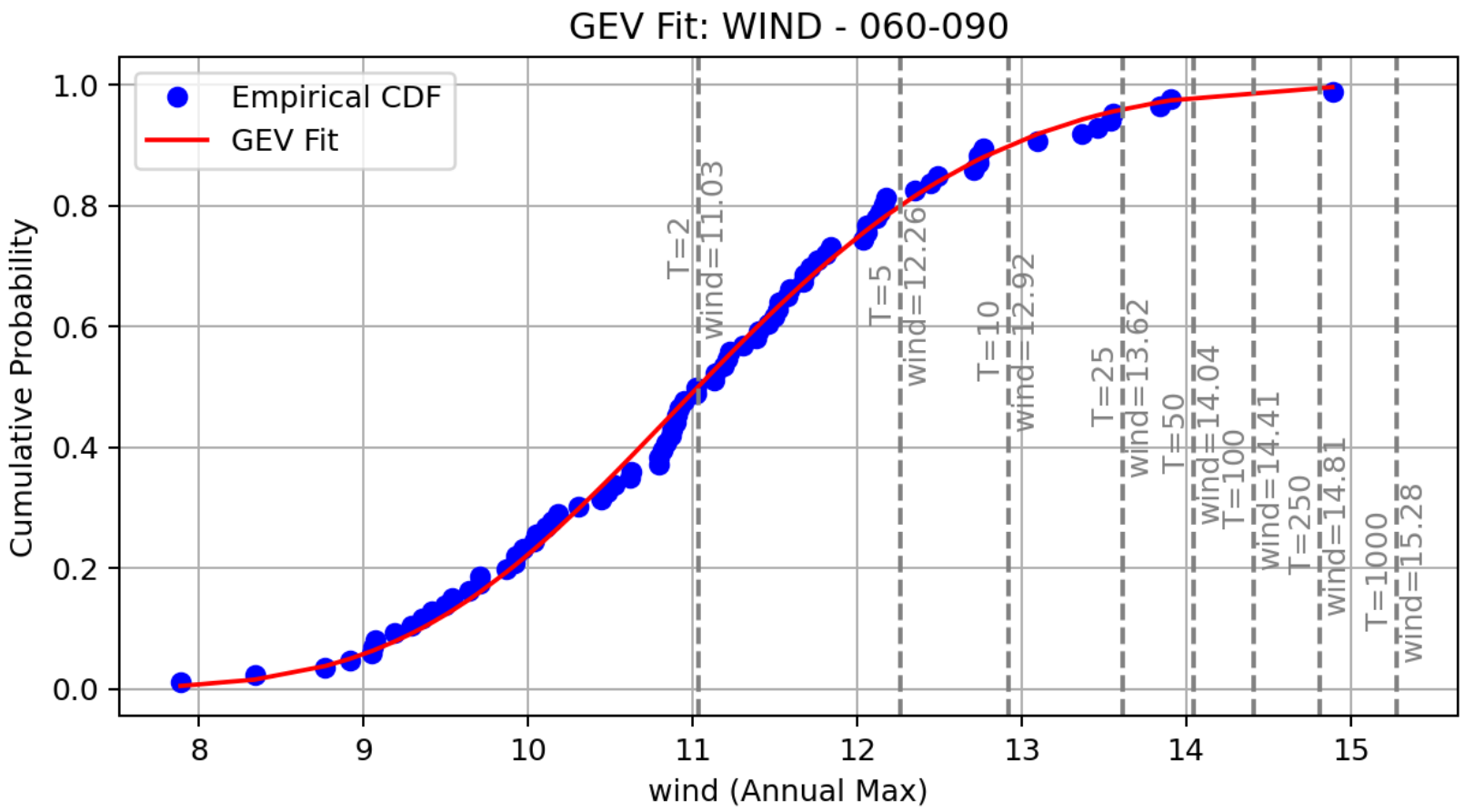
WIND Sector 000-030



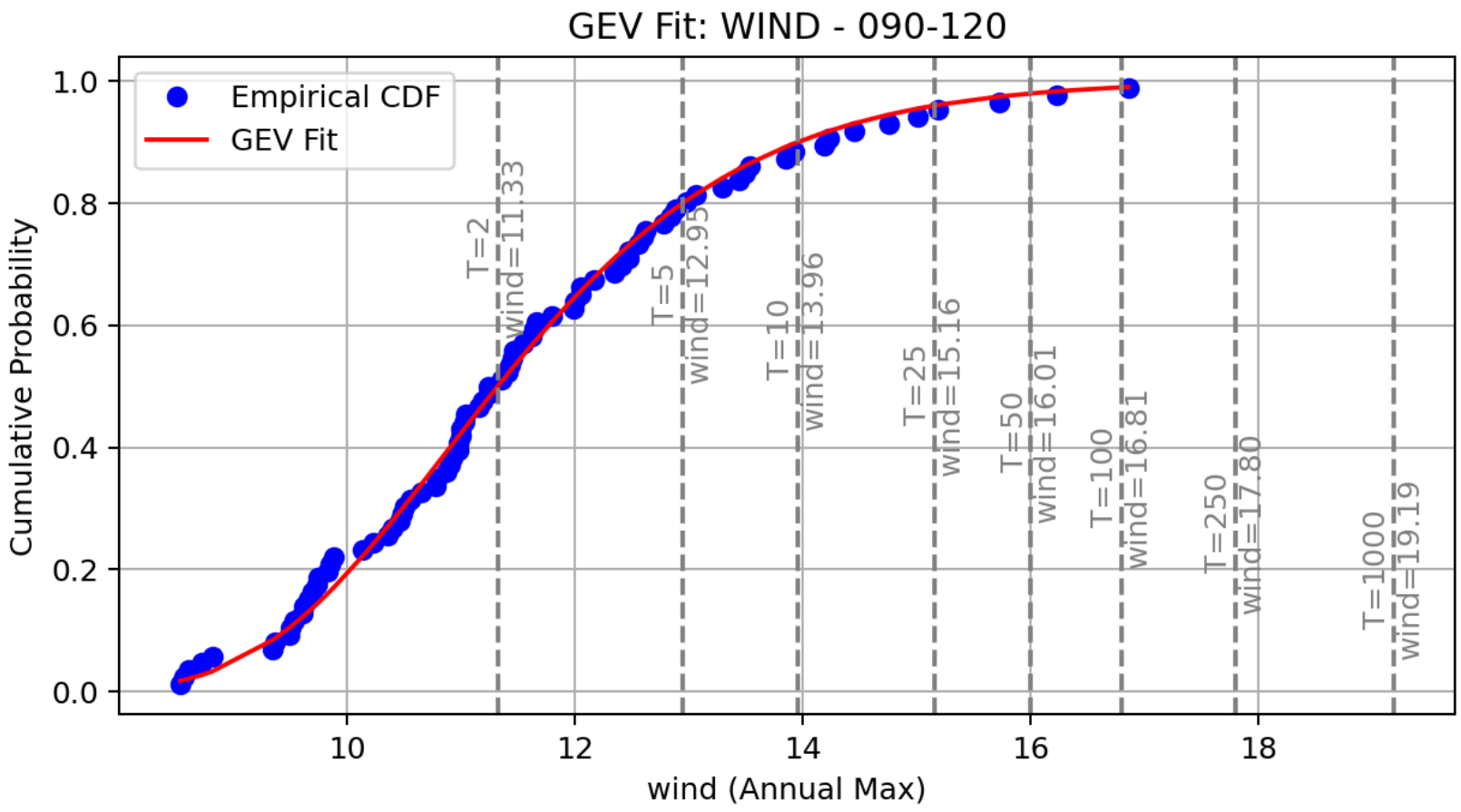
WIND Sector 030-060


GEV Fit: WIND - 030-060
Empirical CDF
GEV Fit
Cumulative Probability
wind (Annual Max)
T=2
wind=13.08
T=5
wind=14.32
T=10
wind=14.95
T=25
wind=15.57
T=50
wind=15.93
T=100
wind=16.22
T=250
wind=16.53
T=1000
wind=16.86
0.0
0.2
0.4
0.6
0.8
1.0
10
11
12
13
14
15
16
17

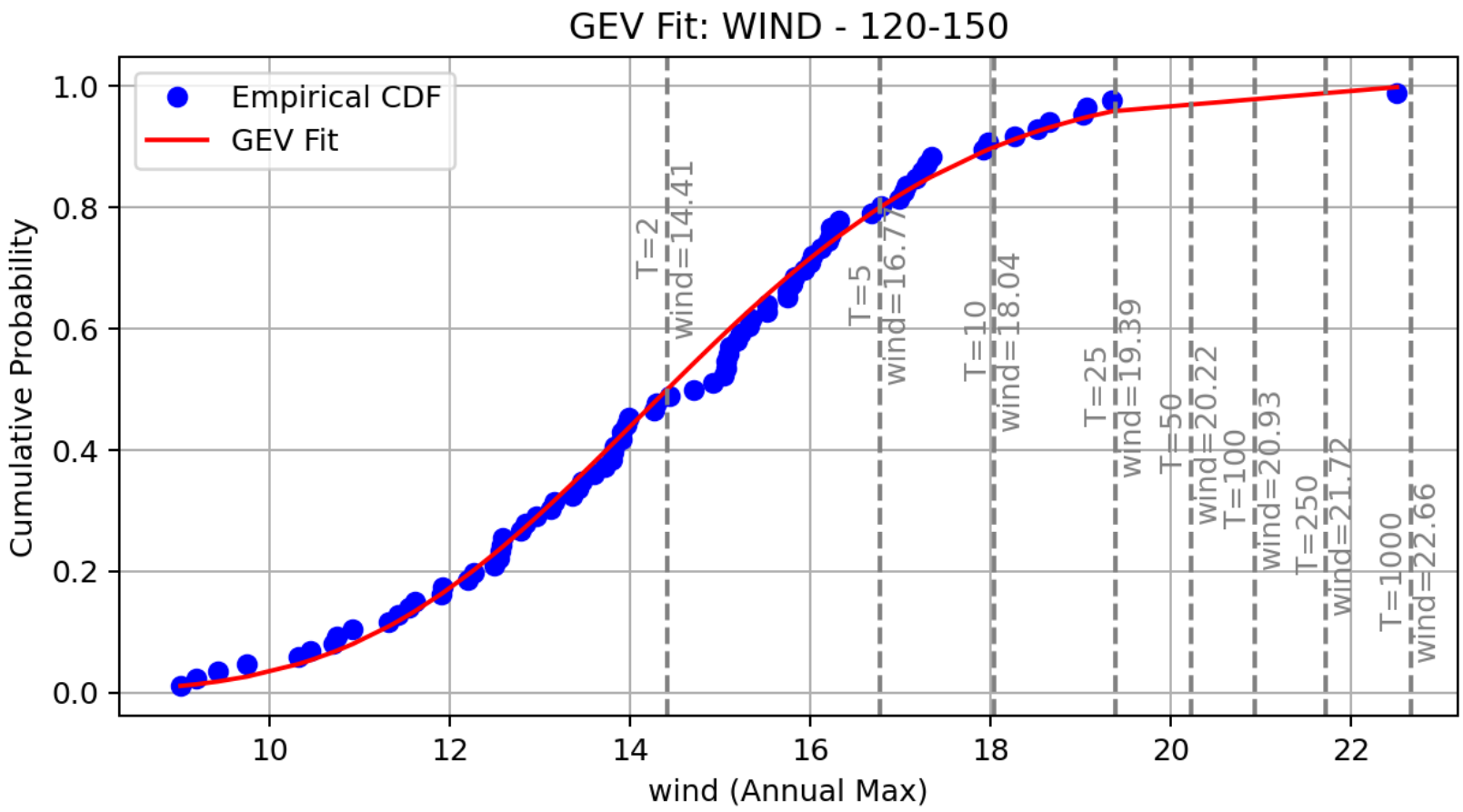
WIND Sector 060-090



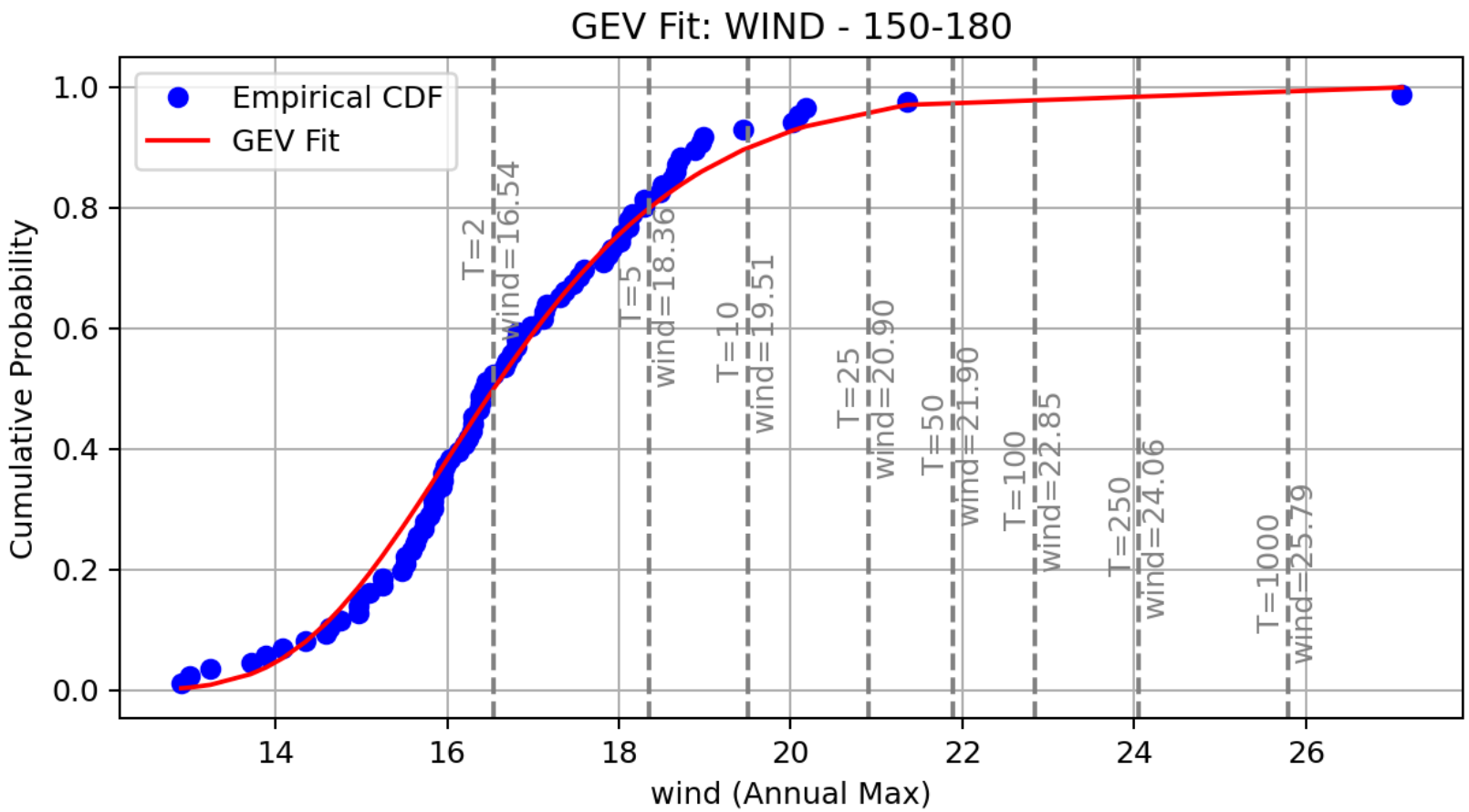
WIND Sector 090-120


GEV Fit: WIND - 090-120
Empirical CDF
GEV Fit
T=2
wind=11.33
T=5
wind=12.95
T=10
wind=13.96
T=25
wind=15.16
T=50
wind=16.01
T=100
wind=16.81
T=250
wind=17.80
T=1000
wind=19.19
Cumulative Probability
wind (Annual Max)
0.0
0.2
0.4
0.6
0.8
1.0
10
12
14
16
18

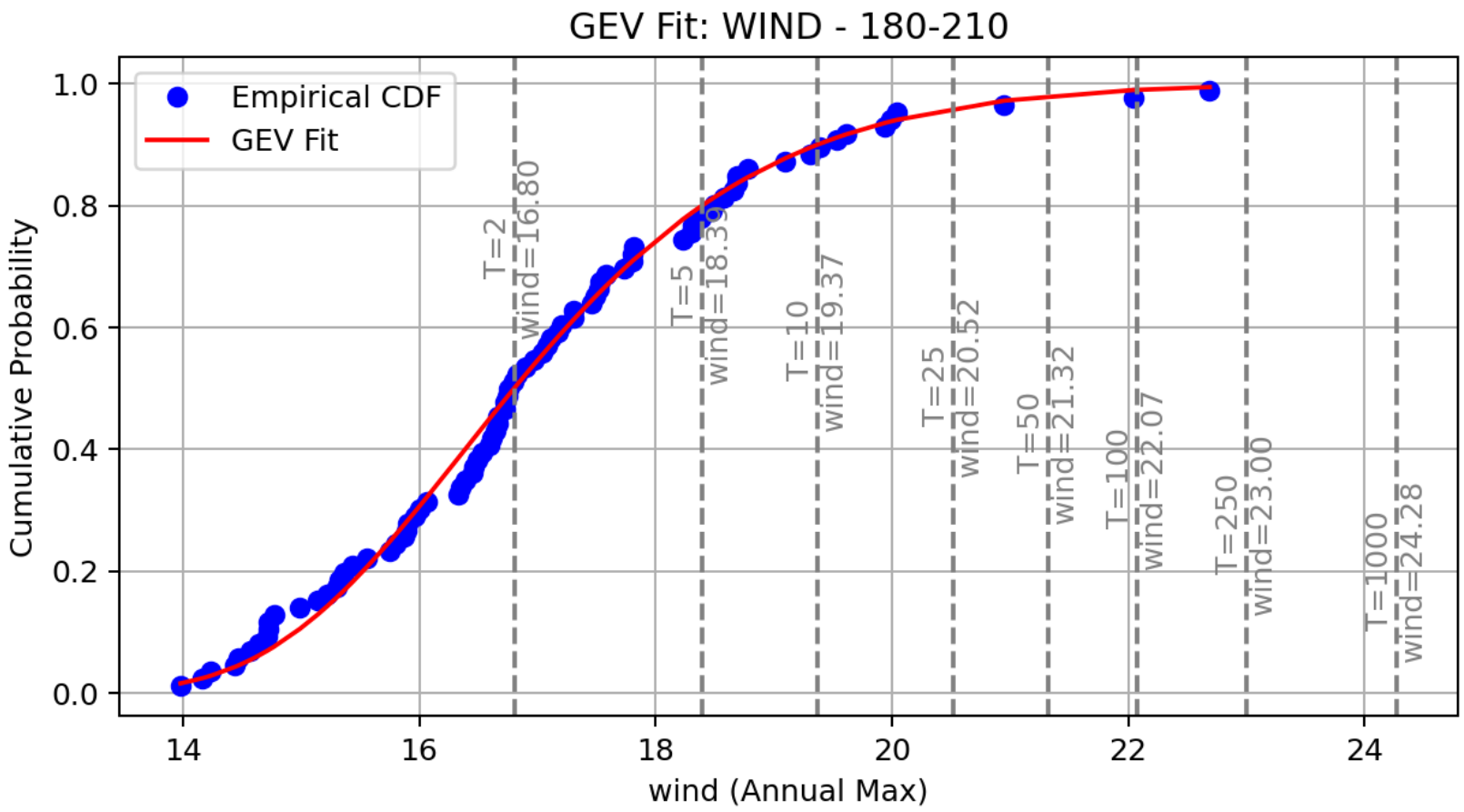
WIND Sector 120-150



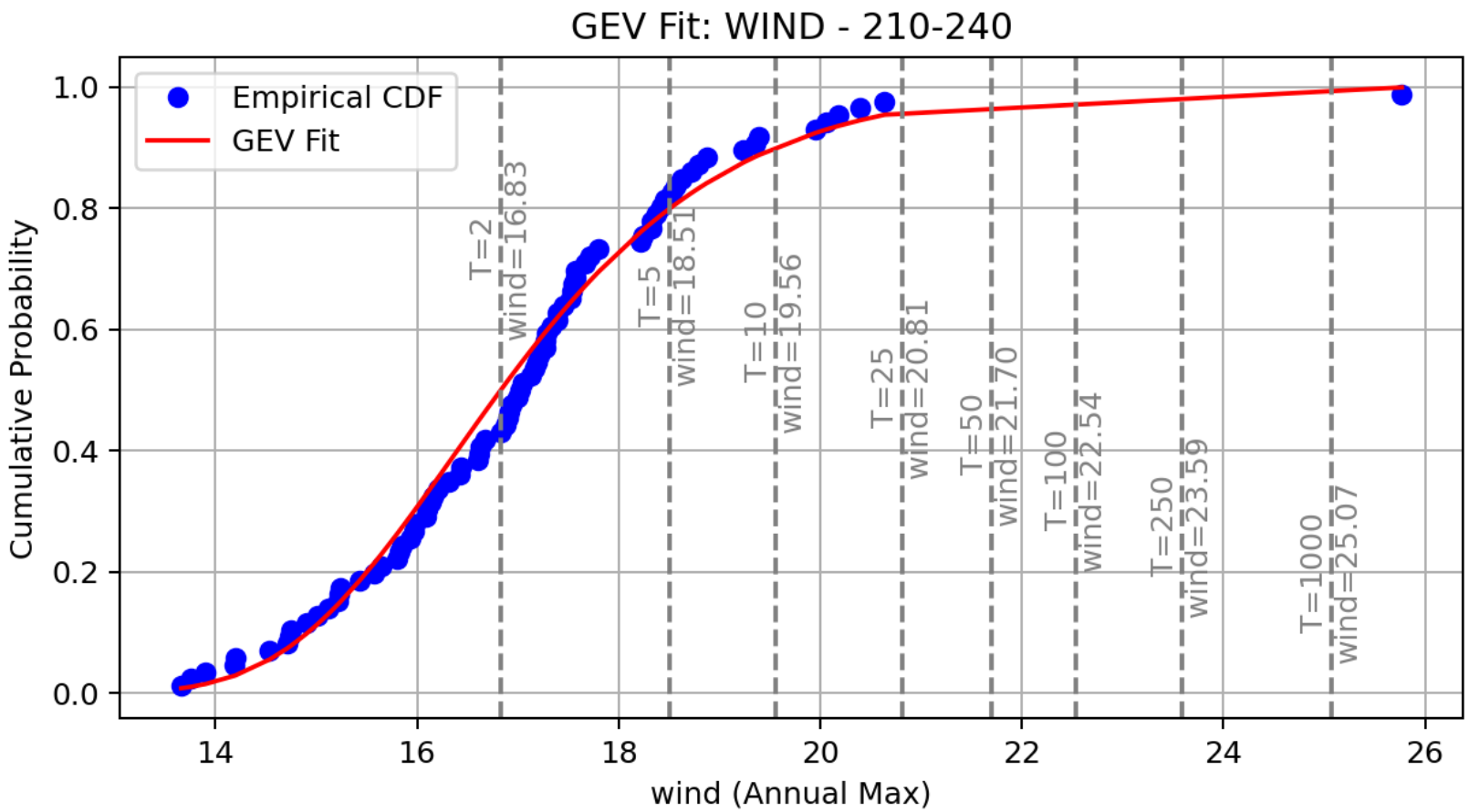
WIND Sector 150-180


GEV Fit: WIND - 150-180
Empirical CDF
GEV Fit
Cumulative Probability
wind (Annual Max)
T=2
wind=16.54
T=5
wind=18.36
T=10
wind=19.51
T=25
wind=20.90
T=50
wind=21.90
T=100
wind=22.85
T=250
wind=24.06
T=1000
wind=25.79
0.0
0.2
0.4
0.6
0.8
1.0
14
16
18
20
22
24
26

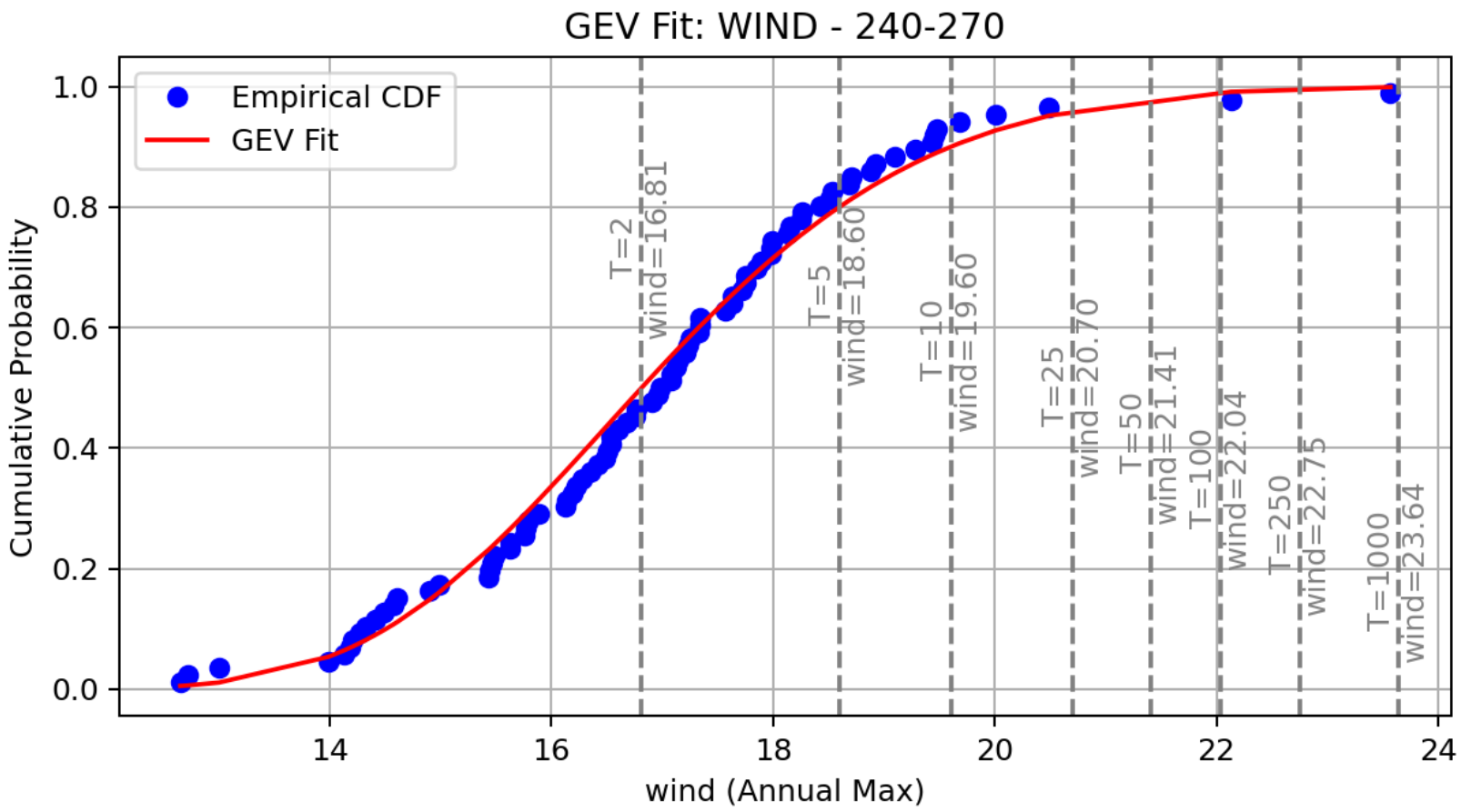
WIND Sector 180-210



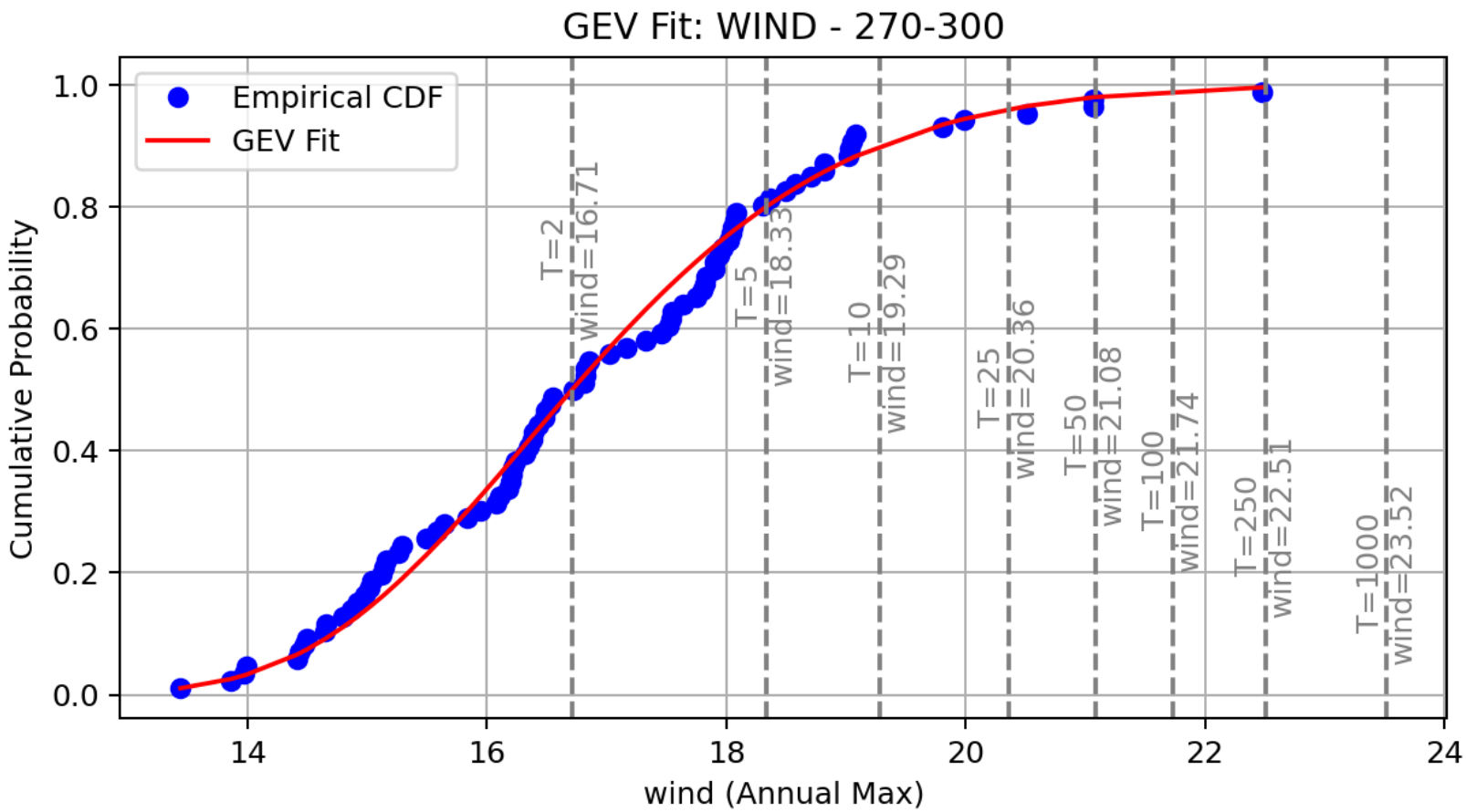
WIND Sector 210-240


GEV Fit: WIND - 210-240
Empirical CDF
GEV Fit
T=2
wind=16.83
T=5
wind=18.51
T=10
wind=19.56
T=25
wind=20.81
T=50
wind=21.70
T=100
wind=22.54
T=250
wind=23.59
T=1000
wind=25.07
Cumulative Probability
wind (Annual Max)
0.0
0.2
0.4
0.6
0.8
1.0
14
16
18
20
22
24
26

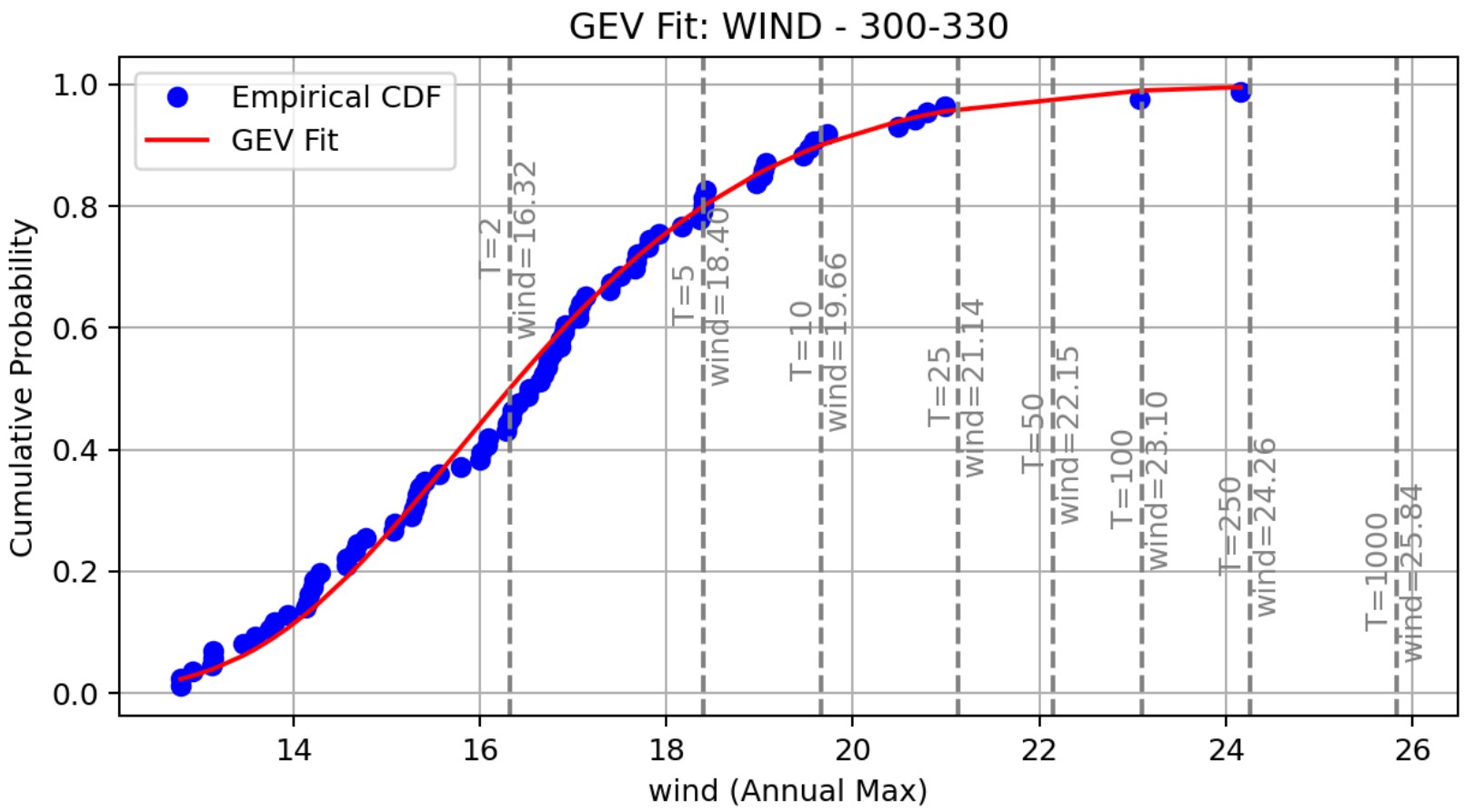
WIND Sector 240-270



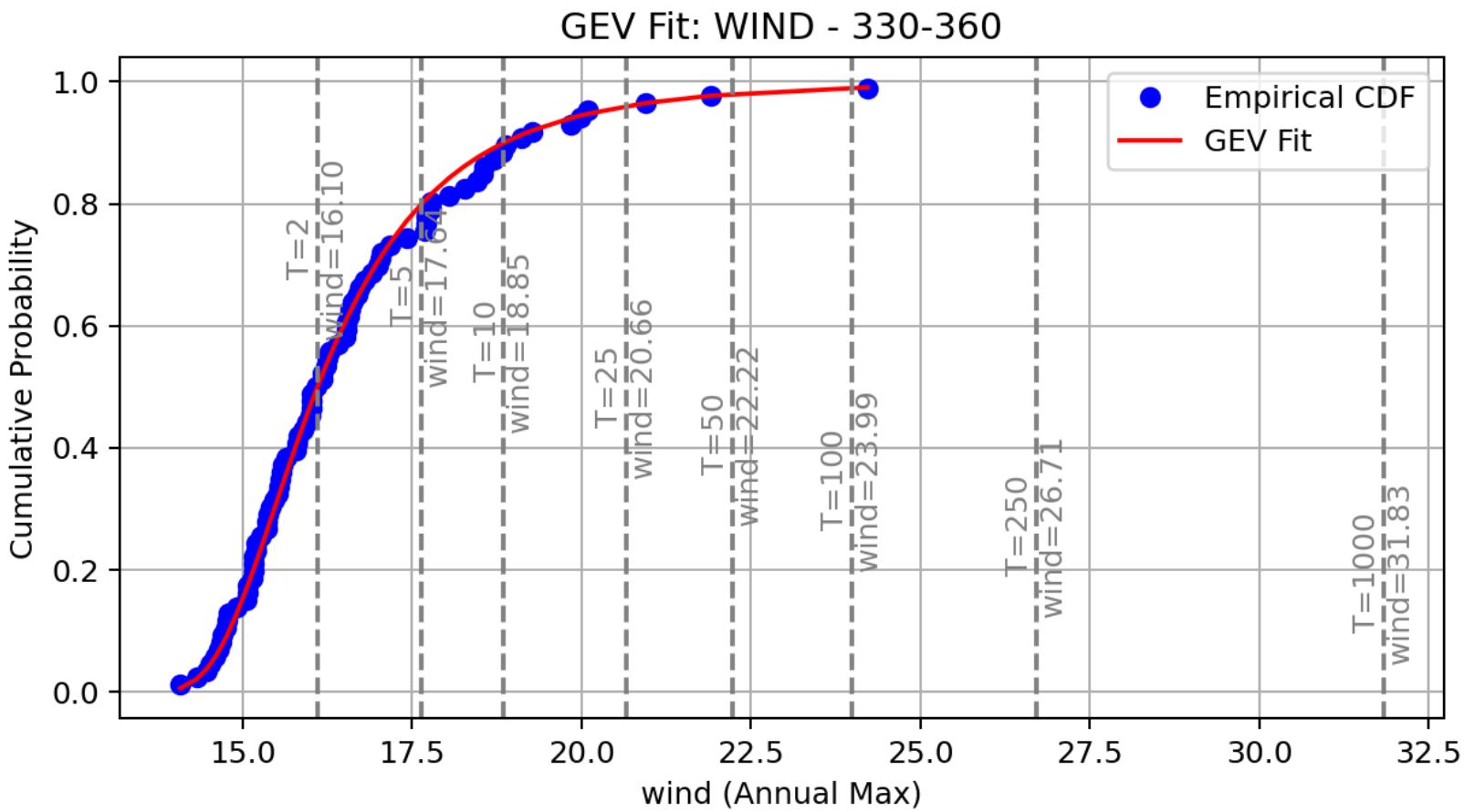
WIND Sector 270-300


GEV Fit: WIND - 270-300
Empirical CDF
GEV Fit
Cumulative Probability
wind (Annual Max)
T=2
wind=16.71
T=5
wind=18.33
T=10
wind=19.29
T=25
wind=20.36
T=50
wind=21.08
T=100
wind=21.74
T=250
wind=22.51
T=1000
wind=23.52
0.0
0.2
0.4
0.6
0.8
1.0
14
16
18
20
22
24

WIND Sector 300-330



WIND Sector 330-360


GEV Fit: WIND - 330-360
Empirical CDF
GEV Fit
T=2
wind=16.10
T=5
wind=17.64
T=10
wind=18.85
T=25
wind=20.66
T=50
wind=22.22
T=100
wind=23.99
T=250
wind=26.71
T=1000
wind=31.83
Cumulative Probability
wind (Annual Max)
0.0
0.2
0.4
0.6
0.8
1.0
15.0
17.5
20.0
22.5
25.0
27.5
30.0
32.5

6) Joint Dist: swh vs mwd (%)

Ind...	(-0.00...	(120.0...	(150.0...	(180.0...	(210.0...	(240.0...	(270.0...	(30.0...	(300.0...	(330.0...	(60.0...	(90.0,...
(0.409, 1.488]	0.37	0.02	0.02	0.11	0.39	1.45	6.13	0.05	11.11	6.12	0.02	0.01
(1.488, 2.566]	0.87	0.06	0.13	0.42	1.08	2.77	9.63	0.08	15.67	13.43	0.04	0.03
(10.112, 11.19]	0	0	0	0.00	0.00	0.00	0.00	0	0.00	0	0	0
(2.566, 3.644]	0.42	0.01	0.08	0.30	0.78	1.91	5.23	0.00	6.95	4.34	0.00	0.00
(3.644, 4.722]	0.05	0.00	0.03	0.11	0.35	0.86	2.35	0	2.39	0.78	0.00	0.00
(4.722, 5.8]	0.00	0	0.01	0.02	0.08	0.31	0.80	0	0.80	0.13	0	0
(5.8, 6.878]	0	0	0.00	0.00	0.02	0.08	0.24	0	0.26	0.02	0	0
(6.878, 7.956]	0	0	0.00	0	0.01	0.02	0.08	0	0.08	0.00	0	0
(7.956, 9.034]	0	0	0.00	0	0	0.01	0.02	0	0.02	0.00	0	0
(9.034, 10.112]	0	0	0	0	0	0.00	0.00	0	0.00	0	0	0

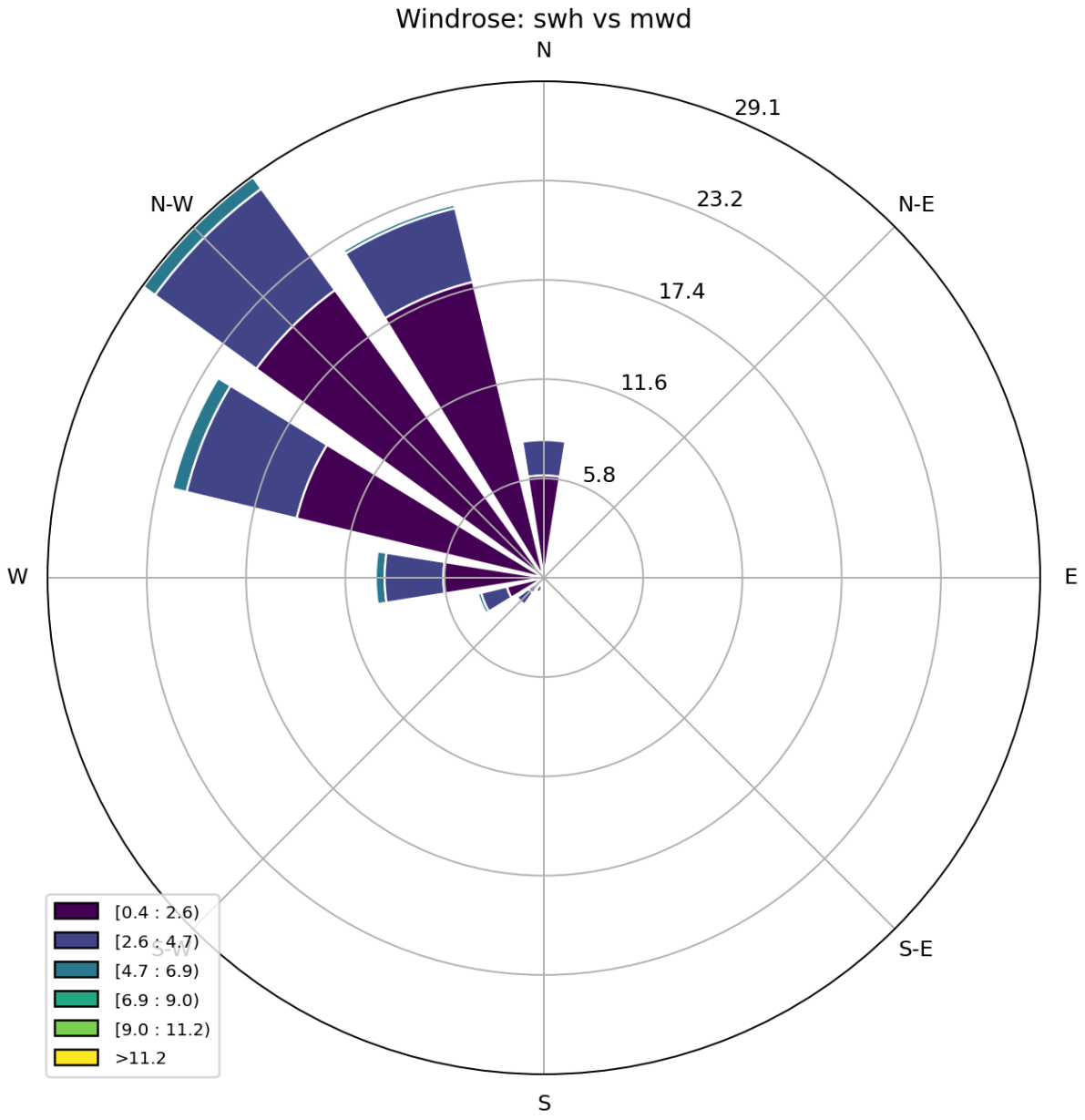
Joint Dist: swh vs pp1d (%)

Index	(10.36,...	(12.2, ...	(14.04, ...	(15.88, ...	(17.72, ...	(19.56,...	(2.999,...	(4.84,...	(6.68,...	(8.52, ...
(0.409,	1.488] 5.47	1.33	0.36	0.10	0.01	0	0.30	2.10	6.89	9.24
(1.488,	2.566]14.92	7.75	1.91	0.43	0.05	0.00	0.02	3.70	6.84	8.58
(10.112,	11.19] 0	0.00	0.00	0.00	0	0	0	0	0	0
(2.566,	3.644] 5.14	6.89	2.90	0.50	0.05	0.00	0	0.09	2.68	1.78
(3.644,	4.722] 1.20	2.33	2.02	0.58	0.05	0.00	0	0	0.19	0.55
(4.722,	5.8] 0.27	0.60	0.72	0.42	0.05	0.00	0	0	0.01	0.08
(5.8, 6.	878] 0.05	0.14	0.21	0.19	0.03	0.00	0	0	0	0.01
(6.878,	7.956] 0.01	0.04	0.07	0.07	0.01	0.00	0	0	0	0.00
(7.956,	9.034] 0.00	0.01	0.02	0.02	0.00	0	0	0	0	0
(9.034,	10.112] 0	0.00	0.00	0.00	0.00	0	0	0	0	0

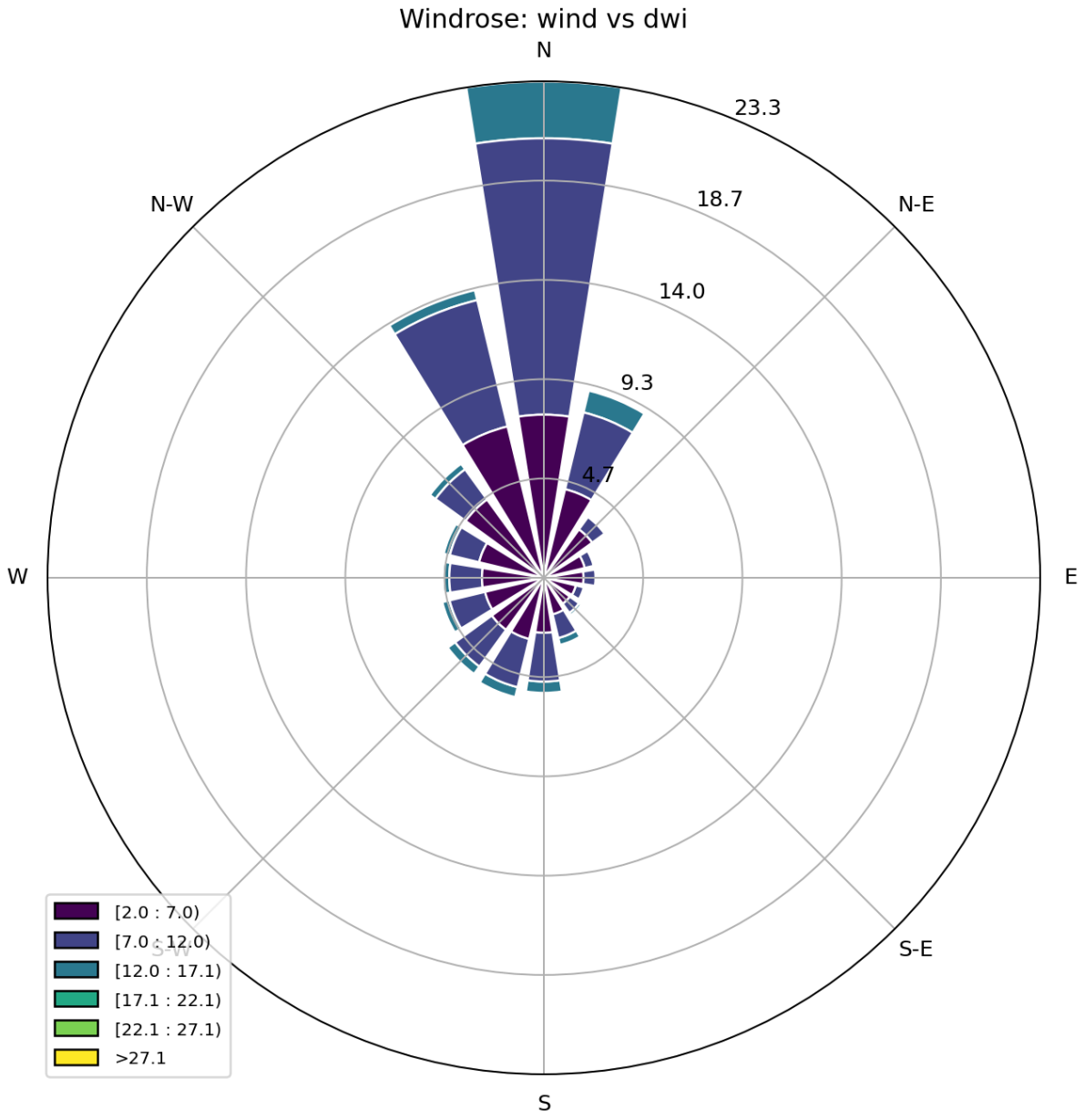
Joint Dist: wind vs dwi (%)

Ind...	(-0.00...	(120.0...	(150.0...	(180.0...	(210.0...	(240.0...	(270.0...	(30.0...	(300.0...	(330.0...	(60.0...	(90.0,...
(1.999, 4.512]	90	1.26	1.44	1.86	2.12	2.20	2.51	2.03	3.29	3.66	1.53	1.36
(12.048, 14.56]	20	0.13	0.46	0.52	0.45	0.36	0.30	0.06	0.33	1.50	0.01	0.02
(14.56, 17.072]	27	0.04	0.13	0.14	0.11	0.09	0.09	0.00	0.09	0.14	0.00	0.00
(17.072, 19.584]	1	0.01	0.01	0.02	0.02	0.02	0.01	0	0.02	0.01	0	0.00
(19.584, 22.096]	0	0.00	0.00	0.00	0.00	0.00	0.00	0	0.00	0.00	0	0
(22.096, 24.608]	0	0.00	0	0.00	0.00	0.00	0.00	0	0.00	0.00	0	0
(24.608, 27.12]	0	0.00	0.00	0	0.00	0	0	0	0	0	0	0
(4.512, 7.024]	36	0.71	1.29	2.02	1.84	1.61	1.56	1.63	2.80	6.77	1.04	0.87
(7.024, 9.536]	63	0.46	1.19	1.93	1.67	1.35	1.21	0.74	1.58	7.94	0.55	0.45
(9.536, 12.048]	30	0.28	0.91	1.28	1.10	0.86	0.72	0.28	0.81	5.27	0.12	0.13

7) Windrose: swh vs mwd


Windrose: swh vs mwd
N
N-E
E
S-E
S
S-W
W
N-W
5.8
11.6
17.4
23.2
29.1
[0.4 : 2.6)
[2.6 : 4.7)
[4.7 : 6.9)
[6.9 : 9.0)
[9.0 : 11.2)
>11.2

8) Windrose: wind vs dwi


Windrose: wind vs dwi
N
N-E
E
S-E
S
S-W
W
N-W
23.3
18.7
14.0
9.3
4.7
[2.0 : 7.0)
[7.0 : 12.0)
[12.0 : 17.1)
[17.1 : 22.1)
[22.1 : 27.1)
>27.1